# PM2.5 Forecasting in Open-Pit Mining: Horizon-Dependent Predictive Value of Local Monitoring and Regional Atmospheric Information

## Supplementary Methods

This supplement documents the mechanics behind the frozen parameters, criteria, and decisions reported in Methodology Sections 2.1–2.12.

*S2.3 MERRA-2 temporal-alignment sensitivity scenarios (supports Section 2.3)*

Three frozen sensitivity scenarios were retained alongside the primary area-weighted CST hour-end alignment: (i) nearest-cell spatial extraction with CST hour-end timing, which changed spatial support only; (ii) area-weighted extraction with CST hour-start labeling, a retrospective timing sensitivity in which the associated interval ended after the nominal label; and (iii) nearest-cell extraction with CST hour-start labeling, which changed both spatial and temporal assumptions. The hour-start scenarios matched 17,536 of 17,544 monitoring timestamps, leaving eight boundary records unmatched (versus nine under the primary hour-end convention). Scenario roles were frozen before fitting and were not selected using $R^2$, RMSE, MAE, test results, or SHAP values.

*S2.4 Leakage-control verification (supports Section 2.4)*

The leakage-control design combined five safeguards: target-time partition assignment; training-only estimation of caps and imputation values; uncapped held-out responses; backward-looking environmental features; and exclusion of future or target-time MERRA-2 measurements. Input identities, feature counts, timestamp continuity, split boundaries, and the absence of explicit future MERRA-2 fields were checked using machine-readable assertions before any model was fit.

Ordinary monitoring-variable missingness was measured before imputation. Missing counts were PM2.5 587, PM10 605, SO2 604, CO 594, NO2 592, O3 592, wind direction 55, wind speed 11, precipitation 10, temperature 3, relative humidity 3, and pressure 0. For every variable with a raw missingness indicator in the feature contract, that indicator was calculated from the unfilled source field and retained after the predictor itself was imputed. Thus imputation did not erase information about whether an observation had originally been unavailable.

The cap-then-median sequence first created the fixed training-calendar mask. The 99.5th-percentile cap for each designated pollutant was then estimated from nonmissing training observations and applied to a predictor-only copy across all partitions. Each capped predictor alias or uncapped direct predictor was subsequently selected as the imputation input, and its median was estimated on training rows only and used unchanged to fill missing values in training, validation, and test predictor inputs. Validation or test observations never contributed to a cap or median.

| Predictor | Training-derived 99.5th-percentile cap | Validation values above cap | Test values above cap |
|---|---|---|---|
| PM2.5 | 223.070 | 0 | 2 |
| PM10 | 352.108 | 26 | 0 |
| SO2 | 121.260 | 3 | 0 |
| CO | 2.620 | 1 | 4 |

The MERRA-2 archive itself contained complete numerical values for all 731 daily subsets. Temporal alignment left nine monitoring timestamps without a matched MERRA-2 interval at the record boundary; these direct MERRA-2 inputs were filled with training-only medians, while merra2_origin_available preserved their unavailable status. After feature construction, history-edge NaNs from lag and rolling operations were left for CatBoost's native missing-value handling rather than filled with future or backcast values. Derived ratios or indicators were likewise permitted to remain NaN when their denominator or physical validity condition was not met.

Pressure had no formally missing source values, but three observations were recorded as 0.0 kPa and were physically invalid. In the executed pipeline, the ordinary model_input_pressure series retained those zeros; direct pressure lags, its 24 h rolling mean, the pressure–wind interaction, and winter/heating interactions therefore inherited the unfiltered values. A separate pressure_clean series masked values outside 80–110 kPa only for the high-pressure stagnation flag and 6/24 h pressure-tendency indicators. This produced exactly 12 non-edge tendency-indicator NaNs, which CatBoost handled natively. The asymmetric treatment is documented to reproduce the archived analysis and is not presented as an ideal physical-data cleaning rule.

A post-hoc validation-only sensitivity converted pressure outside 80–110 kPa to missing before the existing training-only median calculation and before any feature construction. It rebuilt both the archived-pressure and corrected-pressure inputs only through 7 August 2021 18:00, then refitted both model specifications at h1–h24 using the archived horizon-specific CatBoost parameters, original training targets, and original validation targets. Ninety-six fits were completed. The latest raw, target, and prediction timestamp equaled the validation endpoint; locked-test predictions and metrics remained untouched, and their SHA-256 hashes were unchanged before and after execution. Full results are provided in Supplementary Table S12.

PM2.5 targets were never imputed. A row was excluded whenever its horizon-specific target was missing or lay beyond the observed record. The 587 missing PM2.5 hours therefore removed $587 \times 24 = 14{,}088$ in-record origin–horizon pairs. The record endpoint removed another $1 + 2 + \ldots + 24 = 300$ pairs. Consequently, $17{,}544 \times 24 - 14{,}088 - 300 = 406{,}668$ eligible pairs remained across h1–h24.

*S2.5 Full predictor construction rules (supports Section 2.5)*

PM2.5 and PM10 histories included current values; 1, 3, 6, 12, and 24 h lags; missingness indicators; and 6 and 24 h rolling summaries (means, maxima, and standard deviations for PM2.5; means and maxima for PM10). NO2, SO2, and CO included current values, 1, 6, and 24 h lags, missingness indicators, and 24 h means. O3, temperature, relative humidity, pressure, and wind speed used current values, 1, 6, and 24 h lags, missingness indicators where defined, and 24 h means. Additional combinations included the PM2.5/PM10 ratio, PM10 minus PM2.5, the sum of NO2 and SO2, temperature × relative humidity, and pressure × wind speed.

Origin-time calendar variables included cyclic hour, day-of-year, and month encodings; month and season; winter, heating-season, and cold-season indicators; and winter-progress terms (winter = December–February, heating season = November–March, cold season = October–March). Selected local measurements and interaction terms were multiplied by winter and heating-season indicators to allow seasonally varying relationships.

The 23 engineered local-condition indicators were mechanistically motivated transformations of local variables only: wind direction sine/cosine components; calm, low, moderate, and high wind regimes; 6 h wind tendency; low-ventilation and stagnation conditions; 6 and 24 h pressure tendencies; humidity interactions with PM2.5, PM10, and temperature; current and accumulated precipitation; a recent wet-suppression indicator; the PM10/PM2.5 ratio; and coarse-particle indicators (PM10 minus PM2.5).

The MERRA-2 block's 73 predictors comprised, for each of the eight selected aerosol variables, current, 1, 6, and 24 h lagged values and a 24 h rolling mean (40 fields), one shared availability indicator (41st direct field), and current/rolling interactions with winter and heating-season indicators (32 seasonal-interaction fields). No MERRA-2 interaction was classified as a calendar feature. Nine target-calendar features were known deterministically at

93 forecast issuance: sine/cosine encodings of target hour, target day of year, and target month;
94 numerical target day of year; and target winter/heating-season indicators.

*Supplementary Table S1. MERRA-2 aerosol variables retained and their use in the forecasting models.*

| MERRA-2 code | Atmospheric information represented | Unit | Predictors constructed for modeling | Intended predictive context |
|---|---|---|---|---|
| DUCMASS | Total dust aerosol column mass density | kg m−2 | Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values | Regional total mineral-dust loading |
| DUCMASS25 | Dust PM2.5 column mass density | kg m−2 | Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values | Regional fine-dust burden relevant to PM2.5 |
| DUEXTTAU | Dust extinction aerosol optical thickness at 550 nm | Dimensionless | Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values | Optical loading associated with total dust |
| DUEXTT25 | Dust-PM2.5 extinction aerosol optical thickness at 550 nm | Dimensionless | Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values | Optical loading associated with fine dust |
| TOTEXTTAU | Total aerosol extinction optical thickness at 550 nm | Dimensionless | Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values | Overall regional aerosol loading |
| SO4CMASS | Sulfate aerosol column mass density | kg m−2 | Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values | Regional secondary inorganic aerosol burden |
| BCCMASS | Black-carbon aerosol column mass density | kg m−2 | Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values | Regional combustion-related aerosol burden |
| OCCMASS | Organic-carbon aerosol column mass density | kg m−2 | Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values | Regional carbonaceous aerosol burden |

*S2.6 Hyperparameter search space (supports Section 2.6)*

Persistence used the observed origin-time PM2.5 directly and therefore had no fitted hyperparameters. Month–hour climatology used training-period mean PM2.5 by target month and hour, with training-month and global-mean fallbacks. The fixed-parameter local-only and full-model CatBoost validation benchmarks used 400 iterations, depth 6, learning rate 0.05, L2 leaf regularization 5, RMSE loss, random seed 42, and all available processing threads.

The validation-only algorithm sensitivities used method-specific pipelines rather than CatBoost parameters. Ridge regression used median imputation, standardization, and alpha=10.0. Random Forest used median imputation, 60 trees, maximum depth 10, minimum leaf size 4, max_features=0.7, all available processing cores, and seed 42. Histogram gradient boosting used median imputation, 120 iterations, learning rate 0.06, 31 maximum leaf nodes, L2 regularization 2.0, and seed 42. These models were screening comparisons; the accepted tuned CatBoost specification remained fixed.

For the primary CatBoost models, the Optuna search space comprised 250–1,000 iterations, tree depths of 4–8, learning rates from 0.015 to 0.18 (logarithmic scale), L2 leaf regularization from 1 to 25 (logarithmic scale), random strength from 0 to 5, bagging temperature from 0 to 5, and border counts from 64 to 254. The objective was validation RMSE against uncapped PM2.5, with early stopping at 50 rounds per trial; the final iteration count was set to the best iteration plus one, with a minimum of 50. Optuna 4.9.0 used 20 startup trials and then adaptive TPE with deterministic per-search sampler seeds. Each of the 48 horizon-specification studies completed 50 trials, totaling 2,400 validation-only trials. The procedure does not establish globally optimal settings.

*S2.8 Seasonal rolling-origin fold boundaries (supports Section 2.8)*

The spring fold used data through 31 January 2021 for preprocessing training, February 2021 for validation, and March–May 2021 as the held-out test. The summer fold used data through 30 April 2021 for training, May for validation, and June–August for testing. The autumn fold used data through 31 July 2021 for training, August for validation, and September–November for testing. For each fold, predictor caps and imputation medians were re-estimated using that fold's own training history.

Complete winter coverage required two chronologically ordered folds. The 2020–2021 winter fold used January–November 2020 for training, December 2020 for validation, and January–February 2021 for testing. The 2021–2022 winter fold used data through October 2021 for training, November 2021 for validation, and December 2021 for testing. The accepted December predictions were reused without refitting, and out-of-fold January–February and December observations were concatenated (1,410 + 732 = 2,142 test targets per horizon); composite-winter $R^2$, RMSE, MAE, and bias were calculated directly from the pooled observed–predicted pairs rather than by averaging fold-specific $R^2$ values.

*S2.10 Block-length estimation for the moving-block bootstrap (supports Section 2.10)*

After predictions from the frozen models were available but before interval resampling, candidate block lengths were estimated using the positive-sequence integrated autocorrelation time of the full-model h24 residual series, accumulating autocorrelations until the first non-positive value (maximum search 168 h). Estimated lengths were 23 h for the main test, 20 h for spring, 28 h for summer, 25 h for autumn, and 11 h for the original winter series; the maximum value, 28 h, was then fixed for all main-test and seasonal interval calculations.

*S2.12 Full feature-reduction procedure (supports Section 2.12)*

The prospectively specified reduction analysis used 2,390 internal-selection observations per horizon (3 January–12 April 2021 after the 48 h embargo). Exact SHAP and 30-repeat, 28 h block-permutation importance were calculated using models fitted on development data with a fixed CatBoost configuration: 400 iterations, depth 6, learning rate 0.05, L2 leaf regularization 5, RMSE loss, and seed 42. For block permutation, the first 2,380 observations were reordered as 85 complete non-overlapping 28 h blocks while preserving order within each block. The final 10 observations remained unpermuted but were retained in every prediction and $R^2$ calculation. SHAP and positive permutation importance were normalized separately by horizon and summarized across the three bands. Separate 90%-cumulative SHAP and permutation collections were combined by union; repeat-specific permutation selection was recomputed 30 times, and features selected in at least 80% of repeats were classified as stable. Grouped permutations applied the same block reordering to all features in an eight-family block.

The subsequent disclosed comparative study audited 13 candidate ranking methods. Mutual information and F-regression represented filter methods; lasso, elastic net, Extra Trees, Random Forest, and native CatBoost represented embedded methods; RFE, RFECV, forward sequential selection, and custom family

elimination represented wrapper methods; and exact SHAP plus block permutation supplied model-specific evidence. Lasso (alpha=0.001) and elastic net (alpha=0.001, l1_ratio=0.5) did not meet the frozen convergence checkpoint and were excluded from all-horizon scaling. The final matrix therefore contained 11 methods: mutual information, F-regression, Extra Trees (250 trees, minimum leaf size 2), Random Forest (250 trees, minimum leaf size 2, max_features=0.7), native CatBoost using the fixed configuration above, RFE, RFECV, forward sequential selection, custom family elimination, exact SHAP, and block permutation. Wrapper methods used correlation-pruned representatives and three-fold expanding chronological validation with a Ridge (alpha=1) estimator; forward selection used a frozen 25-feature target.

Within each method and horizon, negative importances were set to zero for normalization and each method's positive normalized importance summed to one. At each horizon, the recorded normalized scores available for each feature were averaged; features absent from correlation-pruned wrapper inputs had no wrapper entry and were not assigned zero. The 59 highest-ranked features were retained separately at every horizon. Thus the budget was common, but feature identities were horizon specific. Consensus tiers, cumulative-90% candidates, fixed 75/100/125/150 budgets, band-specific candidates, and subsequent correlation-clustering configurations were evaluated separately and did not directly generate the final 59 lists.

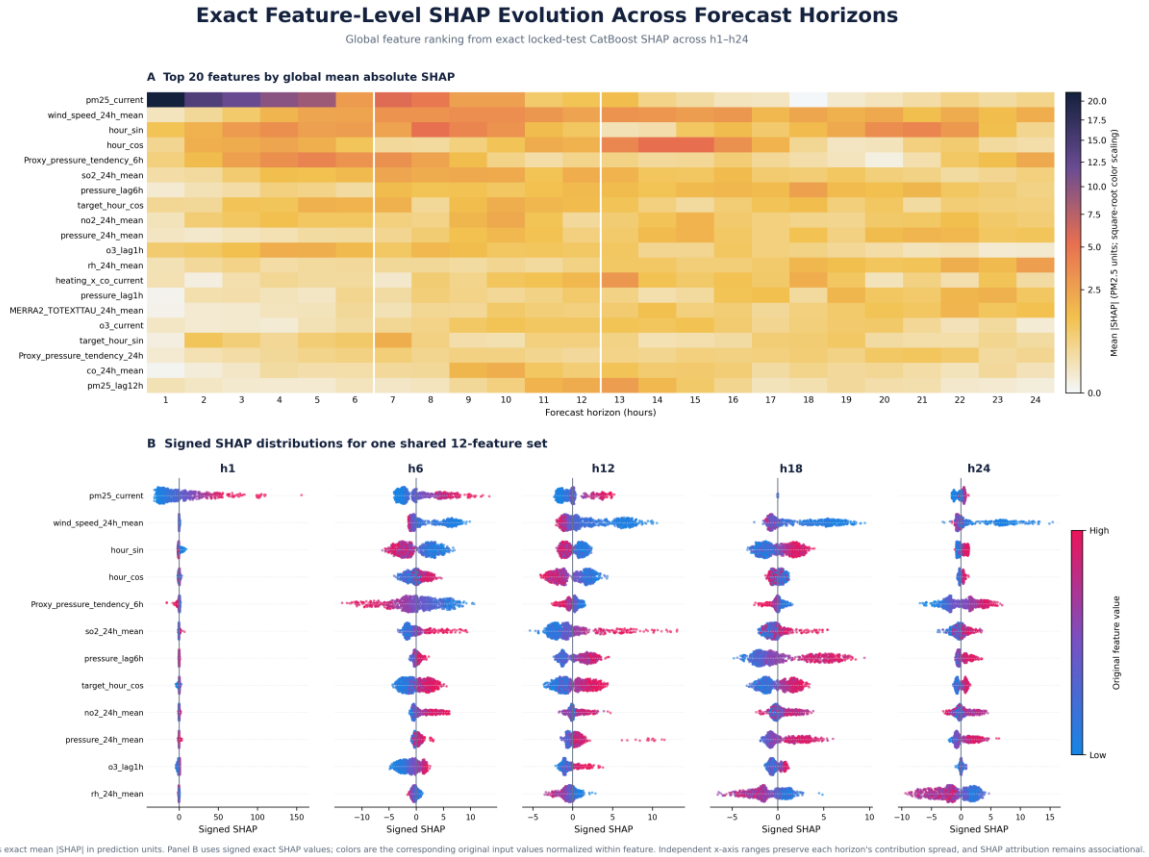
All reduced-versus-full validation refits used the same fixed CatBoost configuration rather than independently tuning each feature count. The post-hoc search evaluated a common 59-predictor budget and band-specific short-, medium-, and long-horizon alternatives under the disclosed joint 5% performance-loss limit. Stability testing used a paired non-circular 28 h moving-block bootstrap with 1,000 resamples drawn from all valid start positions and 30 deterministic refits (seeds 42–71) with only random_seed changed. The same resampled indices were applied to observed, full-model, and reduced-model series. The band-specific

boundary audit compared 14 versus 15 predictors at h6, 50 versus 51 at h10, and 58 versus 59 at h24; the common 59-predictor budget was compared with the full model at h10, h13, and h24. Complete outcomes are reported in Section 3.8.

*Reproducibility (supports Section 2.12)*

Reproducibility was supported by frozen protocol documents, source and output SHA-256 hashes, machine-readable JSON assertions, exact feature lists and family maps, saved Optuna studies and selected parameters, row-level Parquet predictions, preprocessing-parameter tables, fold-boundary tables, complete h1–h24 metrics, and exact SHAP artifacts. The active implementation used Python 3.13.9, pandas 2.3.3, NumPy 2.1.3, CatBoost 1.2.10, Optuna 4.9.0, scikit-learn 1.8.0, SHAP 0.52.0, and PyArrow 21.0.0. Randomized procedures used seed 42 unless a deterministic repeat schedule was explicitly specified. Historical analyses based on an incorrect geographic merge, capped held-out targets, incomplete winter coverage, or superseded event selections were isolated from the active publication package and were not used in this manuscript.

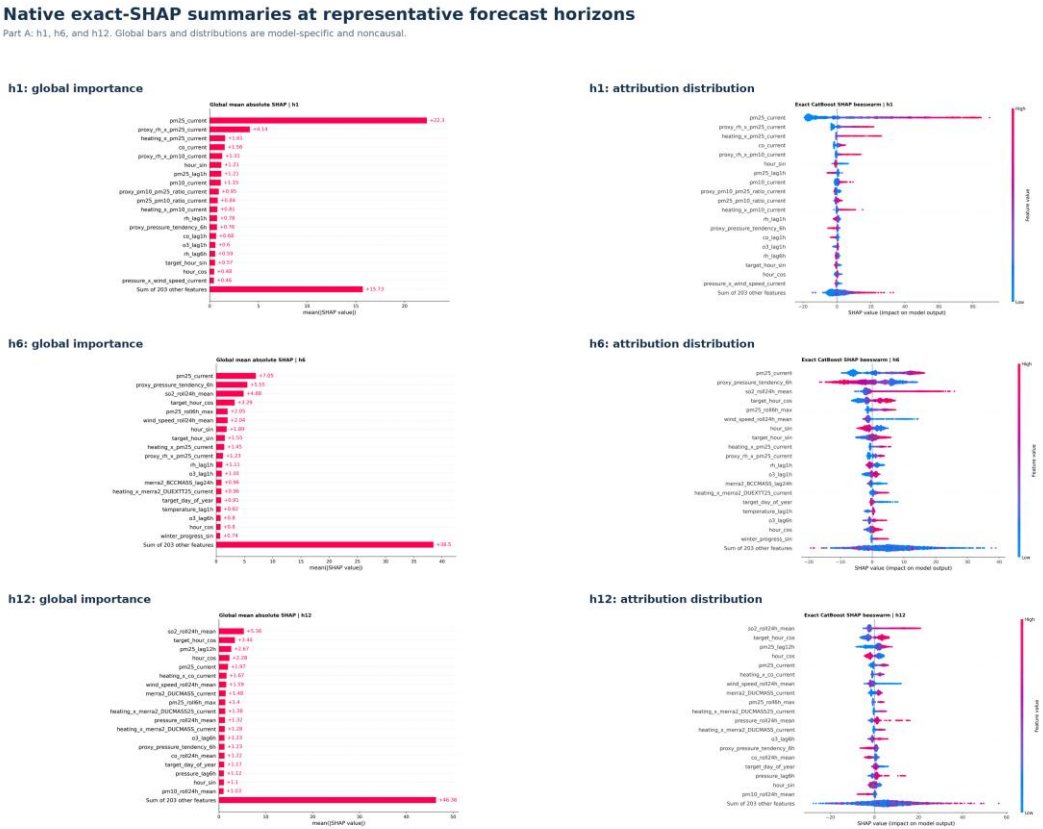
**Supplementary figures**



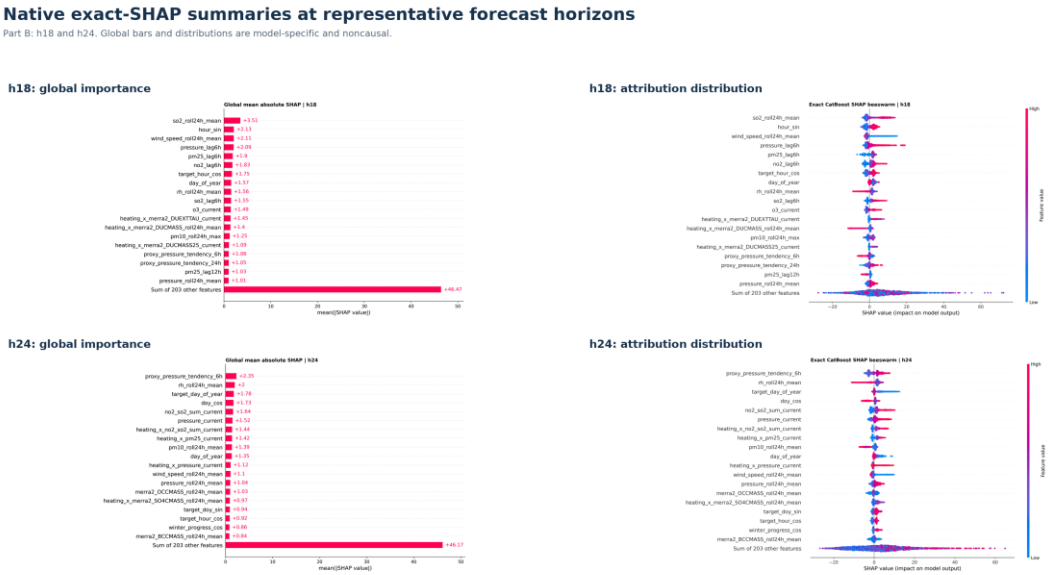

*Supplementary Figure S1. Exact feature-level SHAP evolution across h1–h24 for the primary 222-predictor full models (with MERRA-2). Individual-feature rankings are presented as model attribution rather than causal or independently incremental effects; family-level summaries remain the primary interpretation because correlated predictors may redistribute attribution.*



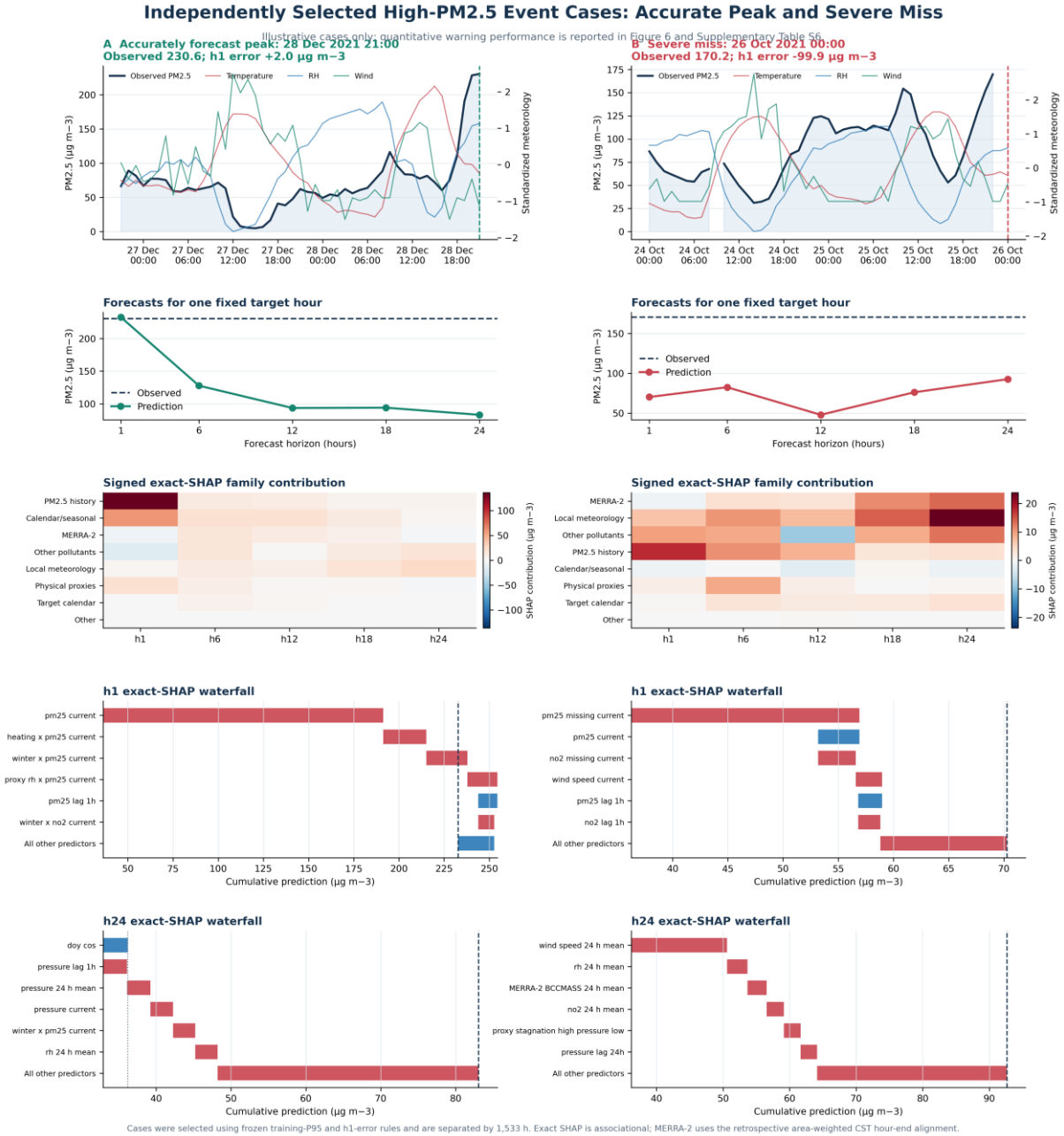

*Supplementary Figure S2. Prespecified validation-only feature-reduction acceptance decision. Horizon-specific reduced model minus full model $R^2$ and relative RMSE and MAE changes are compared with the frozen acceptance limits. For $R^2$, negative values mean the reduced model was below the full model. The individual $R^2$, RMSE, and MAE criteria each passed at 19 of 24 horizons, and 17 horizons passed all three jointly. The reduced version therefore remained supporting validation evidence, and the full 222-predictor model remained primary. Error bars show paired 95% moving-block-bootstrap intervals for reduced model minus full model $R^2$.*




Supplementary Figure S3A. Native exact-SHAP global-importance bars and attribution-distribution plots for h1, h6, and h12. The enlarged panels preserve the native feature-level summaries while improving print readability. These representative summaries complement the family-level main-text analysis; SHAP remains model-specific and noncausal.




Supplementary Figure S3B. Native exact-SHAP global-importance bars and attribution-distribution plots for h18 and h24. The enlarged panels complete the representative horizon sequence shown in Supplementary Figure S3A. SHAP remains model-specific and noncausal.




Cases were selected using frozen training-P95 and h1-error rules and are separated by 1,533 h. Exact SHAP is associational; MERRA-2 uses the retrospective area-weighted CST hour-end alignment.

*Supplementary Figure S4. Independently selected high-PM2.5 event diagnostics for an accurately forecast peak and a severe peak with large forecast error. For each fixed target hour, panels show the preceding 48 h pollution and standardized meteorological context, forecasts at h1/h6/h12/h18/h24, signed family-level exact-SHAP contributions, and h1/h24 exact-SHAP waterfalls. The cases were selected using frozen training-P95 and h1-error rules and are illustrative only; Figure 7 and Supplementary Table S6 provide the quantitative warning evaluation.*

*Supplementary Table S2. Complete locked-test metrics across h1–h24*

All values use the archived primary prediction sets and uncapped observed targets. Persistence has fewer available rows when forecast-origin PM2.5 is missing.

| Model | Horizon | $R^2$ | RMSE (µg m−3) | MAE (µg m−3) | Bias (µg m−3) | n | Predictors |
|---|---|---|---|---|---|---|---|
| Full model (with MERRA-2) | h1 | 0.9488 | 7.019 | 4.157 | 0.415 | 3373 | 222.0 |
| Full model (with MERRA-2) | h2 | 0.8486 | 12.075 | 7.308 | 0.926 | 3373 | 222.0 |
| Full model (with MERRA-2) | h3 | 0.7736 | 14.766 | 9.024 | 1.262 | 3373 | 222.0 |
| Full model (with MERRA-2) | h4 | 0.7071 | 16.793 | 10.532 | 1.362 | 3373 | 222.0 |
| Full model (with MERRA-2) | h5 | 0.6390 | 18.643 | 11.711 | 1.762 | 3373 | 222.0 |
| Full model (with MERRA-2) | h6 | 0.5876 | 19.928 | 12.679 | 1.791 | 3373 | 222.0 |
| Full model (with MERRA-2) | h7 | 0.5446 | 20.941 | 13.354 | 1.608 | 3373 | 222.0 |
| Full model (with MERRA-2) | h8 | 0.4694 | 22.602 | 15.225 | 3.354 | 3373 | 222.0 |
| Full model (with MERRA-2) | h9 | 0.4722 | 22.543 | 14.870 | 2.148 | 3373 | 222.0 |
| Full model (with MERRA-2) | h10 | 0.4361 | 23.301 | 15.536 | 2.149 | 3373 | 222.0 |
| Full model (with MERRA-2) | h11 | 0.4004 | 24.029 | 16.059 | 1.856 | 3373 | 222.0 |

| Model | Horizon | $R^2$ | RMSE (µg m−3) | MAE (µg m−3) | Bias (µg m−3) | n | Predictors |
|---|---|---|---|---|---|---|---|
| Full model (with MERRA-2) | h12 | 0.3809 | 24.416 | 16.529 | 2.350 | 3373 | 222.0 |
| Full model (with MERRA-2) | h13 | 0.3613 | 24.799 | 16.823 | 1.774 | 3373 | 222.0 |
| Full model (with MERRA-2) | h14 | 0.3737 | 24.557 | 16.917 | 2.198 | 3373 | 222.0 |
| Full model (with MERRA-2) | h15 | 0.3252 | 25.490 | 17.316 | 0.912 | 3373 | 222.0 |
| Full model (with MERRA-2) | h16 | 0.3326 | 25.350 | 17.522 | 1.076 | 3373 | 222.0 |
| Full model (with MERRA-2) | h17 | 0.3541 | 24.939 | 17.414 | 1.982 | 3373 | 222.0 |
| Full model (with MERRA-2) | h18 | 0.3388 | 25.233 | 17.781 | 1.900 | 3373 | 222.0 |
| Full model (with MERRA-2) | h19 | 0.3065 | 25.841 | 18.255 | 2.484 | 3373 | 222.0 |
| Full model (with MERRA-2) | h20 | 0.2774 | 26.377 | 18.327 | 1.773 | 3373 | 222.0 |
| Full model (with MERRA-2) | h21 | 0.3146 | 25.690 | 18.041 | 2.608 | 3373 | 222.0 |
| Full model (with MERRA-2) | h22 | 0.2894 | 26.157 | 18.215 | 2.045 | 3373 | 222.0 |
| Full model (with MERRA-2) | h23 | 0.2655 | 26.594 | 18.576 | 2.481 | 3373 | 222.0 |
| Full model (with MERRA-2) | h24 | 0.2044 | 27.677 | 19.284 | 2.086 | 3373 | 222.0 |

| Model | Horizon | $R^2$ | RMSE (µg m−3) | MAE (µg m−3) | Bias (µg m−3) | n | Predictors |
|---|---|---|---|---|---|---|---|
| Local-only model | h1 | 0.9467 | 7.167 | 4.133 | 0.310 | 3373 | 149.0 |
| Local-only model | h2 | 0.8653 | 11.389 | 6.840 | 0.214 | 3373 | 149.0 |
| Local-only model | h3 | 0.7734 | 14.773 | 8.909 | 0.449 | 3373 | 149.0 |
| Local-only model | h4 | 0.7024 | 16.929 | 10.275 | 0.155 | 3373 | 149.0 |
| Local-only model | h5 | 0.6356 | 18.732 | 11.738 | 0.774 | 3373 | 149.0 |
| Local-only model | h6 | 0.5871 | 19.939 | 12.577 | 0.529 | 3373 | 149.0 |
| Local-only model | h7 | 0.5351 | 21.157 | 13.356 | 0.129 | 3373 | 149.0 |
| Local-only model | h8 | 0.4678 | 22.636 | 14.617 | 1.125 | 3373 | 149.0 |
| Local-only model | h9 | 0.4027 | 23.982 | 15.638 | 1.155 | 3373 | 149.0 |
| Local-only model | h10 | 0.3059 | 25.853 | 17.108 | 0.840 | 3373 | 149.0 |
| Local-only model | h11 | 0.3593 | 24.838 | 16.382 | 0.041 | 3373 | 149.0 |
| Local-only model | h12 | 0.3193 | 25.601 | 16.860 | 0.192 | 3373 | 149.0 |
| Local-only model | h13 | 0.2989 | 25.982 | 17.159 | -0.340 | 3373 | 149.0 |

| Model | Horizon | $R^2$ | RMSE (µg m−3) | MAE (µg m−3) | Bias (µg m−3) | n | Predictors |
|---|---|---|---|---|---|---|---|
| Local-only model | h14 | 0.3043 | 25.881 | 17.536 | 0.192 | 3373 | 149.0 |
| Local-only model | h15 | 0.2743 | 26.435 | 17.851 | -0.197 | 3373 | 149.0 |
| Local-only model | h16 | 0.2236 | 27.342 | 18.296 | -0.765 | 3373 | 149.0 |
| Local-only model | h17 | 0.2698 | 26.516 | 17.969 | -0.580 | 3373 | 149.0 |
| Local-only model | h18 | 0.2490 | 26.891 | 18.282 | -0.239 | 3373 | 149.0 |
| Local-only model | h19 | 0.2320 | 27.193 | 18.745 | 0.188 | 3373 | 149.0 |
| Local-only model | h20 | 0.2401 | 27.049 | 18.381 | -0.504 | 3373 | 149.0 |
| Local-only model | h21 | 0.2174 | 27.452 | 18.970 | -0.044 | 3373 | 149.0 |
| Local-only model | h22 | 0.2300 | 27.230 | 18.726 | -0.646 | 3373 | 149.0 |
| Local-only model | h23 | 0.2498 | 26.876 | 18.535 | -0.486 | 3373 | 149.0 |
| Local-only model | h24 | 0.2102 | 27.577 | 19.059 | -0.102 | 3373 | 149.0 |
| Month–hour climatology | h1 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h2 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h3 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h4 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h5 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |

| Model | Horizon | $R^2$ | RMSE (µg m−3) | MAE (µg m−3) | Bias (µg m−3) | n | Predictors |
|---|---|---|---|---|---|---|---|
| Month–hour climatology | h6 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h7 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h8 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h9 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h10 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h11 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h12 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h13 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h14 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h15 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h16 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h17 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h18 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h19 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h20 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h21 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h22 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h23 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Month–hour climatology | h24 | 0.1550 | 28.524 | 20.063 | 1.155 | 3373 | |
| Persistence | h1 | 0.9169 | 8.935 | 4.935 | -0.016 | 3329 | |
| Persistence | h2 | 0.7535 | 15.388 | 8.628 | -0.037 | 3325 | |
| Persistence | h3 | 0.5738 | 20.226 | 11.564 | -0.062 | 3327 | |
| Persistence | h4 | 0.4082 | 23.838 | 13.902 | -0.077 | 3328 | |
| Persistence | h5 | 0.2714 | 26.470 | 15.780 | -0.104 | 3324 | |
| Persistence | h6 | 0.1646 | 28.356 | 17.262 | -0.117 | 3321 | |
| Persistence | h7 | 0.0904 | 29.606 | 18.242 | -0.154 | 3320 | |
| Persistence | h8 | 0.0426 | 30.372 | 18.943 | -0.180 | 3321 | |
| Persistence | h9 | 0.0201 | 30.796 | 19.393 | -0.264 | 3319 | |
| Persistence | h10 | 0.0034 | 31.105 | 19.880 | -0.343 | 3319 | |
| Persistence | h11 | -0.0225 | 31.506 | 20.425 | -0.355 | 3317 | |
| Persistence | h12 | -0.0618 | 32.094 | 20.999 | -0.369 | 3315 | |
| Persistence | h13 | -0.1181 | 32.921 | 21.729 | -0.382 | 3315 | |

| Model | Horizon | $R^2$ | RMSE (µg m−3) | MAE (µg m−3) | Bias (µg m−3) | n | Predictors |
|---|---|---|---|---|---|---|---|
| Persistence | h14 | -0.1869 | 33.907 | 22.466 | -0.421 | 3316 | |
| Persistence | h15 | -0.2571 | 34.889 | 23.149 | -0.444 | 3316 | |
| Persistence | h16 | -0.3157 | 35.722 | 23.658 | -0.486 | 3311 | |
| Persistence | h17 | -0.3639 | 36.357 | 24.021 | -0.446 | 3311 | |
| Persistence | h18 | -0.3690 | 36.394 | 24.022 | -0.477 | 3312 | |
| Persistence | h19 | -0.3547 | 36.159 | 23.805 | -0.466 | 3309 | |
| Persistence | h20 | -0.3219 | 35.718 | 23.409 | -0.479 | 3308 | |
| Persistence | h21 | -0.2782 | 35.117 | 22.948 | -0.466 | 3306 | |
| Persistence | h22 | -0.2291 | 34.432 | 22.418 | -0.467 | 3307 | |
| Persistence | h23 | -0.1938 | 33.945 | 22.102 | -0.456 | 3305 | |
| Persistence | h24 | -0.1930 | 33.788 | 22.163 | -0.424 | 3306 | |

*Supplementary Table S3. Selected CatBoost hyperparameters by model and horizon*

The lowest-validation-RMSE completed trial was selected within each 50-trial seeded study. Each search used 20 startup trials and then the adaptive TPE phase; the locked test was not used for selection.

| Model | Horizon | Iterations | Depth | Learning rate | L2 leaf reg. | Random strength | Bagging temperature | Border count | Seed |
|---|---|---|---|---|---|---|---|---|---|
| Local-only model | h1 | 374 | 4 | 0.06788 | 1.7313 | 0.3253 | 4.7444 | 248 | 42 |
| Full model (with MERRA-2) | h1 | 694 | 4 | 0.06788 | 1.7313 | 0.3253 | 4.7444 | 248 | 42 |
| Local-only model | h2 | 274 | 4 | 0.06788 | 1.7313 | 0.3253 | 4.7444 | 248 | 42 |
| Full model (with MERRA-2) | h2 | 292 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Local-only model | h3 | 329 | 4 | 0.06788 | 1.7313 | 0.3253 | 4.7444 | 248 | 42 |
| Full model (with MERRA-2) | h3 | 254 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Local-only model | h4 | 258 | 7 | 0.08714 | 1.0685 | 4.8495 | 4.1622 | 104 | 42 |
| Full model (with MERRA-2) | h4 | 300 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Local-only model | h5 | 435 | 4 | 0.06788 | 1.7313 | 0.3253 | 4.7444 | 248 | 42 |
| Full model (with MERRA-2) | h5 | 220 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Local-only model | h6 | 420 | 4 | 0.06788 | 1.7313 | 0.3253 | 4.7444 | 248 | 42 |
| Full model (with MERRA-2) | h6 | 319 | 8 | 0.10294 | 20.5761 | 4.4741 | 2.9895 | 240 | 42 |
| Local-only model | h7 | 857 | 5 | 0.01912 | 9.0473 | 2.2008 | 0.6102 | 158 | 42 |
| Full model (with MERRA-2) | h7 | 190 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Local-only model | h8 | 316 | 4 | 0.06788 | 1.7313 | 0.3253 | 4.7444 | 248 | 42 |
| Full model (with MERRA-2) | h8 | 198 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Local-only model | h9 | 206 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Full model (with MERRA-2) | h9 | 508 | 5 | 0.05778 | 1.5740 | 4.0110 | 0.3728 | 252 | 42 |
| Local-only model | h10 | 839 | 5 | 0.01912 | 9.0473 | 2.2008 | 0.6102 | 158 | 42 |
| Full model (with MERRA-2) | h10 | 486 | 5 | 0.05778 | 1.5740 | 4.0110 | 0.3728 | 252 | 42 |
| Local-only model | h11 | 160 | 4 | 0.06788 | 1.7313 | 0.3253 | 4.7444 | 248 | 42 |
| Full model (with MERRA-2) | h11 | 198 | 7 | 0.08714 | 1.0685 | 4.8495 | 4.1622 | 104 | 42 |
| Local-only model | h12 | 121 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Full model (with MERRA-2) | h12 | 183 | 8 | 0.10294 | 20.5761 | 4.4741 | 2.9895 | 240 | 42 |
| Local-only model | h13 | 131 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Full model (with MERRA-2) | h13 | 181 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Local-only model | h14 | 134 | 7 | 0.08714 | 1.0685 | 4.8495 | 4.1622 | 104 | 42 |
| Full model (with MERRA-2) | h14 | 233 | 7 | 0.08714 | 1.0685 | 4.8495 | 4.1622 | 104 | 42 |
| Local-only model | h15 | 310 | 5 | 0.05778 | 1.5740 | 4.0110 | 0.3728 | 252 | 42 |
| Full model (with MERRA-2) | h15 | 229 | 7 | 0.08714 | 1.0685 | 4.8495 | 4.1622 | 104 | 42 |
| Local-only model | h16 | 122 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Full model (with MERRA-2) | h16 | 376 | 8 | 0.10294 | 20.5761 | 4.4741 | 2.9895 | 240 | 42 |
| Local-only model | h17 | 163 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Full model (with MERRA-2) | h17 | 193 | 8 | 0.10294 | 20.5761 | 4.4741 | 2.9895 | 240 | 42 |
| Local-only model | h18 | 186 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Full model (with MERRA-2) | h18 | 163 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Local-only model | h19 | 254 | 8 | 0.10294 | 20.5761 | 4.4741 | 2.9895 | 240 | 42 |

| Model | Horizon | Iterations | Depth | Learning rate | L2 leaf reg. | Random strength | Bagging temperature | Border count | Seed |
|---|---|---|---|---|---|---|---|---|---|
| Full model (with MERRA-2) | h19 | 287 | 8 | 0.10294 | 20.5761 | 4.4741 | 2.9895 | 240 | 42 |
| Local-only model | h20 | 465 | 4 | 0.06788 | 1.7313 | 0.3253 | 4.7444 | 248 | 42 |
| Full model (with MERRA-2) | h20 | 517 | 5 | 0.05778 | 1.5740 | 4.0110 | 0.3728 | 252 | 42 |
| Local-only model | h21 | 167 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Full model (with MERRA-2) | h21 | 257 | 7 | 0.08714 | 1.0685 | 4.8495 | 4.1622 | 104 | 42 |
| Local-only model | h22 | 218 | 8 | 0.10294 | 20.5761 | 4.4741 | 2.9895 | 240 | 42 |
| Full model (with MERRA-2) | h22 | 513 | 5 | 0.05778 | 1.5740 | 4.0110 | 0.3728 | 252 | 42 |
| Local-only model | h23 | 442 | 5 | 0.05778 | 1.5740 | 4.0110 | 0.3728 | 252 | 42 |
| Full model (with MERRA-2) | h23 | 268 | 8 | 0.10294 | 20.5761 | 4.4741 | 2.9895 | 240 | 42 |
| Local-only model | h24 | 131 | 8 | 0.09248 | 6.8689 | 0.7801 | 0.7800 | 75 | 42 |
| Full model (with MERRA-2) | h24 | 517 | 5 | 0.05778 | 1.5740 | 4.0110 | 0.3728 | 252 | 42 |

*Supplementary Table S4. Complete predictor names, families, and availability rules*

The 222-predictor contract contains only forecast-origin information or deterministic target-calendar fields known at issuance. Draft 3 selected the primary models with 50 validation-only Optuna trials per search.

| Predictor | Family | Availability rule |
|---|---|---|
| hour_sin | Calendar/seasonal timing | forecast origin t or earlier |
| hour_cos | Calendar/seasonal timing | forecast origin t or earlier |
| doy_sin | Calendar/seasonal timing | forecast origin t or earlier |
| doy_cos | Calendar/seasonal timing | forecast origin t or earlier |
| month | Calendar/seasonal timing | forecast origin t or earlier |
| season | Calendar/seasonal timing | forecast origin t or earlier |
| pm25_current | PM2.5 history | forecast origin t or earlier |
| pm25_lag1h | PM2.5 history | forecast origin t or earlier |
| pm25_lag3h | PM2.5 history | forecast origin t or earlier |
| pm25_lag6h | PM2.5 history | forecast origin t or earlier |
| pm25_lag12h | PM2.5 history | forecast origin t or earlier |
| pm25_lag24h | PM2.5 history | forecast origin t or earlier |
| pm25_missing_current | PM2.5 history | forecast origin t or earlier |
| pm25_roll6h_mean | PM2.5 history | forecast origin t or earlier |
| pm25_roll6h_max | PM2.5 history | forecast origin t or earlier |
| pm25_roll6h_std | PM2.5 history | forecast origin t or earlier |
| pm25_roll24h_mean | PM2.5 history | forecast origin t or earlier |
| pm25_roll24h_max | PM2.5 history | forecast origin t or earlier |
| pm25_roll24h_std | PM2.5 history | forecast origin t or earlier |
| pm10_current | Other pollutant history | forecast origin t or earlier |
| pm10_lag1h | Other pollutant history | forecast origin t or earlier |
| pm10_lag3h | Other pollutant history | forecast origin t or earlier |
| pm10_lag6h | Other pollutant history | forecast origin t or earlier |
| pm10_lag12h | Other pollutant history | forecast origin t or earlier |
| pm10_lag24h | Other pollutant history | forecast origin t or earlier |
| pm10_missing_current | Other pollutant history | forecast origin t or earlier |
| pm10_roll6h_mean | Other pollutant history | forecast origin t or earlier |
| pm10_roll6h_max | Other pollutant history | forecast origin t or earlier |
| pm10_roll24h_mean | Other pollutant history | forecast origin t or earlier |
| pm10_roll24h_max | Other pollutant history | forecast origin t or earlier |
| no2_current | Other pollutant history | forecast origin t or earlier |
| no2_lag1h | Other pollutant history | forecast origin t or earlier |
| no2_lag6h | Other pollutant history | forecast origin t or earlier |
| no2_lag24h | Other pollutant history | forecast origin t or earlier |
| no2_missing_current | Other pollutant history | forecast origin t or earlier |
| no2_roll24h_mean | Other pollutant history | forecast origin t or earlier |
| so2_current | Other pollutant history | forecast origin t or earlier |

| Predictor | Family | Availability rule |
|---|---|---|
| so2_lag1h | Other pollutant history | forecast origin t or earlier |
| so2_lag6h | Other pollutant history | forecast origin t or earlier |
| so2_lag24h | Other pollutant history | forecast origin t or earlier |
| so2_missing_current | Other pollutant history | forecast origin t or earlier |
| so2_roll24h_mean | Other pollutant history | forecast origin t or earlier |
| co_current | Other pollutant history | forecast origin t or earlier |
| co_lag1h | Other pollutant history | forecast origin t or earlier |
| co_lag6h | Other pollutant history | forecast origin t or earlier |
| co_lag24h | Other pollutant history | forecast origin t or earlier |
| co_missing_current | Other pollutant history | forecast origin t or earlier |
| co_roll24h_mean | Other pollutant history | forecast origin t or earlier |
| temperature_current | Local meteorology | forecast origin t or earlier |
| temperature_lag1h | Local meteorology | forecast origin t or earlier |
| temperature_lag6h | Local meteorology | forecast origin t or earlier |
| temperature_lag24h | Local meteorology | forecast origin t or earlier |
| temperature_missing_current | Local meteorology | forecast origin t or earlier |
| temperature_roll24h_mean | Local meteorology | forecast origin t or earlier |
| o3_current | Other pollutant history | forecast origin t or earlier |
| o3_lag1h | Other pollutant history | forecast origin t or earlier |
| o3_lag6h | Other pollutant history | forecast origin t or earlier |
| o3_lag24h | Other pollutant history | forecast origin t or earlier |
| o3_missing_current | Other pollutant history | forecast origin t or earlier |
| o3_roll24h_mean | Other pollutant history | forecast origin t or earlier |
| rh_current | Local meteorology | forecast origin t or earlier |
| rh_lag1h | Local meteorology | forecast origin t or earlier |
| rh_lag6h | Local meteorology | forecast origin t or earlier |
| rh_lag24h | Local meteorology | forecast origin t or earlier |
| rh_missing_current | Local meteorology | forecast origin t or earlier |
| rh_roll24h_mean | Local meteorology | forecast origin t or earlier |
| pressure_current | Local meteorology | forecast origin t or earlier |
| pressure_lag1h | Local meteorology | forecast origin t or earlier |
| pressure_lag6h | Local meteorology | forecast origin t or earlier |
| pressure_lag24h | Local meteorology | forecast origin t or earlier |
| pressure_missing_current | Local meteorology | forecast origin t or earlier |
| pressure_roll24h_mean | Local meteorology | forecast origin t or earlier |
| wind_speed_current | Local meteorology | forecast origin t or earlier |
| wind_speed_lag1h | Local meteorology | forecast origin t or earlier |
| wind_speed_lag6h | Local meteorology | forecast origin t or earlier |
| wind_speed_lag24h | Local meteorology | forecast origin t or earlier |
| wind_speed_missing_current | Local meteorology | forecast origin t or earlier |
| wind_speed_roll24h_mean | Local meteorology | forecast origin t or earlier |
| pm25_pm10_ratio_current | PM2.5 history | forecast origin t or earlier |
| pm10_minus_pm25_current | Other pollutant history | forecast origin t or earlier |
| no2_so2_sum_current | Other pollutant history | forecast origin t or earlier |
| temp_x_rh_current | Local meteorology | forecast origin t or earlier |

| Predictor | Family | Availability rule |
|---|---|---|
| pressure_x_wind_speed_current | Local meteorology | forecast origin t or earlier |
| day_of_year | Calendar/seasonal timing | forecast origin t or earlier |
| month_sin | Calendar/seasonal timing | forecast origin t or earlier |
| month_cos | Calendar/seasonal timing | forecast origin t or earlier |
| is_winter | Calendar/seasonal timing | forecast origin t or earlier |
| is_heating_season | Calendar/seasonal timing | forecast origin t or earlier |
| is_cold_season | Other | forecast origin t or earlier |
| winter_progress_sin | Calendar/seasonal timing | forecast origin t or earlier |
| winter_progress_cos | Calendar/seasonal timing | forecast origin t or earlier |
| winter_x_pm25_current | Calendar/seasonal timing | forecast origin t or earlier |
| heating_x_pm25_current | Calendar/seasonal timing | forecast origin t or earlier |
| winter_x_pm25_roll24h_mean | Calendar/seasonal timing | forecast origin t or earlier |
| heating_x_pm25_roll24h_mean | Calendar/seasonal timing | forecast origin t or earlier |
| winter_x_pm10_current | Calendar/seasonal timing | forecast origin t or earlier |
| heating_x_pm10_current | Calendar/seasonal timing | forecast origin t or earlier |
| winter_x_no2_current | Calendar/seasonal timing | forecast origin t or earlier |
| heating_x_no2_current | Calendar/seasonal timing | forecast origin t or earlier |
| winter_x_so2_current | Calendar/seasonal timing | forecast origin t or earlier |
| heating_x_so2_current | Calendar/seasonal timing | forecast origin t or earlier |
| winter_x_co_current | Calendar/seasonal timing | forecast origin t or earlier |
| heating_x_co_current | Calendar/seasonal timing | forecast origin t or earlier |
| winter_x_temperature_current | Local meteorology | forecast origin t or earlier |
| heating_x_temperature_current | Local meteorology | forecast origin t or earlier |
| winter_x_rh_current | Local meteorology | forecast origin t or earlier |
| heating_x_rh_current | Local meteorology | forecast origin t or earlier |
| winter_x_pressure_current | Local meteorology | forecast origin t or earlier |
| heating_x_pressure_current | Local meteorology | forecast origin t or earlier |
| winter_x_wind_speed_current | Local meteorology | forecast origin t or earlier |
| heating_x_wind_speed_current | Local meteorology | forecast origin t or earlier |
| winter_x_o3_current | Calendar/seasonal timing | forecast origin t or earlier |
| heating_x_o3_current | Calendar/seasonal timing | forecast origin t or earlier |
| winter_x_temp_x_rh_current | Local meteorology | forecast origin t or earlier |
| heating_x_temp_x_rh_current | Local meteorology | forecast origin t or earlier |
| winter_x_no2_so2_sum_current | Calendar/seasonal timing | forecast origin t or earlier |
| heating_x_no2_so2_sum_current | Calendar/seasonal timing | forecast origin t or earlier |
| proxy_wind_dir_sin_current | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_wind_dir_cos_current | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_wind_calm | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_wind_low | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_wind_moderate | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_wind_high | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_wind_speed_tendency_6h | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_ventilation_wind_only | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_ventilation_low_flag | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_stagnation_high_pressure_low_wind | Engineered local-condition indicators | forecast origin t or earlier |

| Predictor | Family | Availability rule |
|---|---|---|
| proxy_pressure_tendency_6h | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_pressure_tendency_24h | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_rh_x_pm10_current | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_rh_x_pm25_current | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_rh_x_temperature_current | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_precip_current | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_precip_any_current | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_precip_roll6h_sum | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_precip_roll24h_sum | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_recent_wet_suppression_flag | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_pm10_pm25_ratio_current | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_pmcoarse_current | Engineered local-condition indicators | forecast origin t or earlier |
| proxy_pmcoarse_floor0_current | Engineered local-condition indicators | forecast origin t or earlier |
| merra2_origin_available | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUCMASS_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUCMASS_lag1h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUCMASS_lag6h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUCMASS_lag24h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUCMASS_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUCMASS25_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUCMASS25_lag1h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUCMASS25_lag6h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUCMASS25_lag24h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUCMASS25_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUEXTTAU_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUEXTTAU_lag1h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUEXTTAU_lag6h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUEXTTAU_lag24h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUEXTTAU_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUEXTT25_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUEXTT25_lag1h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUEXTT25_lag6h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUEXTT25_lag24h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_DUEXTT25_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_TOTEXTTAU_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_TOTEXTTAU_lag1h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_TOTEXTTAU_lag6h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_TOTEXTTAU_lag24h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_TOTEXTTAU_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_SO4CMASS_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_SO4CMASS_lag1h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_SO4CMASS_lag6h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_SO4CMASS_lag24h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_SO4CMASS_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_BCCMASS_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |

| Predictor | Family | Availability rule |
|---|---|---|
| merra2_BCCMASS_lag1h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_BCCMASS_lag6h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_BCCMASS_lag24h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_BCCMASS_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_OCCMASS_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_OCCMASS_lag1h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_OCCMASS_lag6h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_OCCMASS_lag24h | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| merra2_OCCMASS_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_DUCMASS_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_DUCMASS_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_DUCMASS_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_DUCMASS_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_DUCMASS25_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_DUCMASS25_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_DUCMASS25_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_DUCMASS25_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_DUEXTTAU_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_DUEXTTAU_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_DUEXTTAU_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_DUEXTTAU_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_DUEXTT25_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_DUEXTT25_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_DUEXTT25_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_DUEXTT25_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_TOTEXTTAU_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_TOTEXTTAU_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_TOTEXTTAU_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_TOTEXTTAU_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_SO4CMASS_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_SO4CMASS_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_SO4CMASS_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_SO4CMASS_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_BCCMASS_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_BCCMASS_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_BCCMASS_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_BCCMASS_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_OCCMASS_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_OCCMASS_current | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| winter_x_merra2_OCCMASS_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| heating_x_merra2_OCCMASS_roll24h_mean | MERRA-2 aerosol/reanalysis | forecast origin t or earlier |
| target_hour_sin | Known target-calendar timing | known deterministically at forecast origin |
| target_hour_cos | Known target-calendar timing | known deterministically at forecast origin |
| target_doy_sin | Known target-calendar timing | known deterministically at forecast origin |
| target_doy_cos | Known target-calendar timing | known deterministically at forecast origin |

| Predictor | Family | Availability rule |
|---|---|---|
| target_day_of_year | Known target-calendar timing | known deterministically at forecast origin |
| target_month_sin | Known target-calendar timing | known deterministically at forecast origin |
| target_month_cos | Known target-calendar timing | known deterministically at forecast origin |
| target_is_winter | Known target-calendar timing | known deterministically at forecast origin |
| target_is_heating_season | Known target-calendar timing | known deterministically at forecast origin |

*Supplementary Table S5. Seasonal h1–h24 performance and paired R²-difference intervals*

Spring, summer, and autumn use their rolling-origin folds. Composite winter pools January–February and December 2021 predictions from two chronologically ordered folds; bootstrap blocks never cross the intervening gap.

| Season | Horizon | Local $R^2$ | MERRA-2 $R^2$ | $\Delta R^2$ | 95% CI lower | 95% CI upper | MERRA-2 RMSE (µg m−3) | MERRA-2 MAE (µg m−3) | n |
|---|---|---|---|---|---|---|---|---|---|
| Spring | h1 | 0.9315 | 0.9307 | -0.0008 | -0.0036 | 0.0017 | 9.722 | 5.805 | 2208 |
| Spring | h2 | 0.8295 | 0.8263 | -0.0031 | -0.0080 | 0.0016 | 15.386 | 9.328 | 2208 |
| Spring | h3 | 0.7336 | 0.7408 | 0.0071 | -0.0064 | 0.0181 | 18.798 | 11.723 | 2208 |
| Spring | h4 | 0.6620 | 0.6683 | 0.0063 | -0.0163 | 0.0251 | 21.265 | 13.668 | 2208 |
| Spring | h5 | 0.6040 | 0.6093 | 0.0053 | -0.0213 | 0.0290 | 23.077 | 15.226 | 2208 |
| Spring | h6 | 0.5535 | 0.5492 | -0.0043 | -0.0349 | 0.0287 | 24.788 | 16.574 | 2208 |
| Spring | h7 | 0.4907 | 0.5073 | 0.0166 | -0.0359 | 0.0709 | 25.915 | 17.170 | 2208 |
| Spring | h8 | 0.4839 | 0.4756 | -0.0083 | -0.0454 | 0.0218 | 26.736 | 18.009 | 2208 |
| Spring | h9 | 0.4478 | 0.4460 | -0.0018 | -0.0452 | 0.0448 | 27.481 | 18.661 | 2208 |
| Spring | h10 | 0.4069 | 0.3852 | -0.0217 | -0.0802 | 0.0359 | 28.949 | 19.647 | 2208 |
| Spring | h11 | 0.3814 | 0.3684 | -0.0130 | -0.0739 | 0.0446 | 29.342 | 19.823 | 2208 |
| Spring | h12 | 0.3776 | 0.3649 | -0.0126 | -0.0817 | 0.0503 | 29.422 | 19.854 | 2208 |
| Spring | h13 | 0.3646 | 0.3554 | -0.0092 | -0.0971 | 0.0686 | 29.641 | 20.032 | 2208 |
| Spring | h14 | 0.3745 | 0.3431 | -0.0314 | -0.1090 | 0.0368 | 29.923 | 20.342 | 2208 |
| Spring | h15 | 0.3323 | 0.3409 | 0.0086 | -0.0581 | 0.0656 | 29.974 | 19.917 | 2208 |
| Spring | h16 | 0.3126 | 0.3021 | -0.0105 | -0.0811 | 0.0567 | 30.844 | 20.682 | 2208 |
| Spring | h17 | 0.3165 | 0.3099 | -0.0066 | -0.0990 | 0.0774 | 30.672 | 20.735 | 2208 |
| Spring | h18 | 0.2986 | 0.2924 | -0.0063 | -0.0839 | 0.0714 | 31.058 | 20.564 | 2208 |
| Spring | h19 | 0.2871 | 0.2800 | -0.0071 | -0.0894 | 0.0675 | 31.328 | 20.745 | 2208 |
| Spring | h20 | 0.2643 | 0.2995 | 0.0352 | -0.0576 | 0.1073 | 30.900 | 21.014 | 2208 |
| Spring | h21 | 0.2649 | 0.3051 | 0.0402 | -0.0304 | 0.1086 | 30.777 | 20.524 | 2208 |
| Spring | h22 | 0.2659 | 0.3027 | 0.0368 | -0.0349 | 0.1089 | 30.830 | 20.462 | 2208 |
| Spring | h23 | 0.2875 | 0.3188 | 0.0314 | -0.0235 | 0.0837 | 30.472 | 20.767 | 2208 |
| Spring | h24 | 0.2556 | 0.3071 | 0.0516 | -0.0040 | 0.1098 | 30.732 | 20.599 | 2208 |
| Summer | h1 | 0.9060 | 0.9067 | 0.0007 | -0.0024 | 0.0034 | 4.264 | 3.008 | 2091 |
| Summer | h2 | 0.7767 | 0.7759 | -0.0008 | -0.0080 | 0.0053 | 6.609 | 4.659 | 2091 |
| Summer | h3 | 0.6651 | 0.6702 | 0.0052 | -0.0090 | 0.0183 | 8.018 | 5.757 | 2091 |
| Summer | h4 | 0.5839 | 0.5810 | -0.0029 | -0.0207 | 0.0125 | 9.038 | 6.562 | 2091 |
| Summer | h5 | 0.5087 | 0.5154 | 0.0068 | -0.0141 | 0.0278 | 9.718 | 7.071 | 2091 |
| Summer | h6 | 0.4642 | 0.4670 | 0.0028 | -0.0246 | 0.0298 | 10.193 | 7.368 | 2091 |
| Summer | h7 | 0.4188 | 0.4243 | 0.0055 | -0.0244 | 0.0316 | 10.593 | 7.696 | 2091 |
| Summer | h8 | 0.3963 | 0.3966 | 0.0003 | -0.0455 | 0.0486 | 10.845 | 7.831 | 2091 |
| Summer | h9 | 0.3690 | 0.3857 | 0.0167 | -0.0354 | 0.0575 | 10.943 | 7.941 | 2091 |
| Summer | h10 | 0.3707 | 0.3864 | 0.0157 | -0.0366 | 0.0656 | 10.936 | 8.037 | 2091 |
| Summer | h11 | 0.3461 | 0.3594 | 0.0133 | -0.0287 | 0.0544 | 11.174 | 8.129 | 2091 |
| Summer | h12 | 0.2961 | 0.3536 | 0.0575 | 0.0110 | 0.0988 | 11.225 | 8.161 | 2091 |
| Summer | h13 | 0.2721 | 0.3129 | 0.0407 | 0.0071 | 0.0729 | 11.573 | 8.489 | 2091 |

| Season | Horizon | Local $R^2$ | MERRA-2 $R^2$ | $\Delta R^2$ | 95% CI lower | 95% CI upper | MERRA-2 RMSE (μg m−3) | MERRA-2 MAE (μg m−3) | n |
|---|---|---|---|---|---|---|---|---|---|
| Summer | h14 | 0.2485 | 0.2903 | 0.0418 | 0.0033 | 0.0840 | 11.761 | 8.722 | 2091 |
| Summer | h15 | 0.2496 | 0.2668 | 0.0172 | -0.0198 | 0.0529 | 11.955 | 8.851 | 2091 |
| Summer | h16 | 0.2278 | 0.2602 | 0.0324 | -0.0118 | 0.0777 | 12.008 | 8.854 | 2091 |
| Summer | h17 | 0.1976 | 0.2381 | 0.0405 | 0.0075 | 0.0729 | 12.186 | 9.000 | 2091 |
| Summer | h18 | 0.1965 | 0.2465 | 0.0500 | 0.0143 | 0.0891 | 12.119 | 8.884 | 2091 |
| Summer | h19 | 0.2206 | 0.2438 | 0.0232 | -0.0092 | 0.0571 | 12.141 | 9.014 | 2091 |
| Summer | h20 | 0.2116 | 0.2436 | 0.0320 | -0.0044 | 0.0686 | 12.142 | 9.025 | 2091 |
| Summer | h21 | 0.2297 | 0.2571 | 0.0274 | -0.0233 | 0.0659 | 12.033 | 9.021 | 2091 |
| Summer | h22 | 0.2110 | 0.2449 | 0.0339 | -0.0127 | 0.0806 | 12.132 | 9.057 | 2091 |
| Summer | h23 | 0.2006 | 0.2238 | 0.0232 | -0.0371 | 0.0710 | 12.300 | 9.153 | 2091 |
| Summer | h24 | 0.2166 | 0.2200 | 0.0034 | -0.0583 | 0.0615 | 12.330 | 9.330 | 2091 |
| Autumn | h1 | 0.9490 | 0.9470 | -0.0020 | -0.0041 | 0.0002 | 7.529 | 4.214 | 2163 |
| Autumn | h2 | 0.8691 | 0.8708 | 0.0017 | -0.0068 | 0.0107 | 11.757 | 6.907 | 2163 |
| Autumn | h3 | 0.7911 | 0.7877 | -0.0034 | -0.0159 | 0.0086 | 15.071 | 9.025 | 2163 |
| Autumn | h4 | 0.7179 | 0.7250 | 0.0071 | -0.0121 | 0.0241 | 17.152 | 10.729 | 2163 |
| Autumn | h5 | 0.6656 | 0.6671 | 0.0014 | -0.0316 | 0.0296 | 18.873 | 11.928 | 2163 |
| Autumn | h6 | 0.6224 | 0.6289 | 0.0066 | -0.0260 | 0.0316 | 19.924 | 12.647 | 2163 |
| Autumn | h7 | 0.5799 | 0.5729 | -0.0070 | -0.0498 | 0.0237 | 21.375 | 13.585 | 2163 |
| Autumn | h8 | 0.5251 | 0.5293 | 0.0042 | -0.0488 | 0.0409 | 22.440 | 14.453 | 2163 |
| Autumn | h9 | 0.4641 | 0.4949 | 0.0308 | -0.0457 | 0.0843 | 23.247 | 14.932 | 2163 |
| Autumn | h10 | 0.4246 | 0.4769 | 0.0523 | -0.0033 | 0.0971 | 23.656 | 15.384 | 2163 |
| Autumn | h11 | 0.4103 | 0.4620 | 0.0518 | -0.0373 | 0.1125 | 23.990 | 15.959 | 2163 |
| Autumn | h12 | 0.4142 | 0.4384 | 0.0242 | -0.0732 | 0.1092 | 24.512 | 16.484 | 2163 |
| Autumn | h13 | 0.3675 | 0.4089 | 0.0414 | -0.0436 | 0.1048 | 25.147 | 16.849 | 2163 |
| Autumn | h14 | 0.3221 | 0.3966 | 0.0744 | -0.0195 | 0.1500 | 25.407 | 17.200 | 2163 |
| Autumn | h15 | 0.3024 | 0.3890 | 0.0866 | -0.0139 | 0.1810 | 25.567 | 17.158 | 2163 |
| Autumn | h16 | 0.2949 | 0.3879 | 0.0930 | 0.0157 | 0.1686 | 25.589 | 17.006 | 2163 |
| Autumn | h17 | 0.2770 | 0.3888 | 0.1118 | 0.0124 | 0.2024 | 25.571 | 17.534 | 2163 |
| Autumn | h18 | 0.2596 | 0.3695 | 0.1098 | 0.0149 | 0.2004 | 25.972 | 17.604 | 2163 |
| Autumn | h19 | 0.2532 | 0.3515 | 0.0983 | 0.0070 | 0.1719 | 26.340 | 17.725 | 2163 |
| Autumn | h20 | 0.2813 | 0.3551 | 0.0738 | 0.0116 | 0.1232 | 26.265 | 17.655 | 2163 |
| Autumn | h21 | 0.2640 | 0.3250 | 0.0610 | -0.0126 | 0.1146 | 26.872 | 17.684 | 2163 |
| Autumn | h22 | 0.2583 | 0.3381 | 0.0799 | 0.0178 | 0.1416 | 26.609 | 17.720 | 2163 |
| Autumn | h23 | 0.2786 | 0.3425 | 0.0639 | -0.0295 | 0.1401 | 26.521 | 17.971 | 2163 |
| Autumn | h24 | 0.2396 | 0.3334 | 0.0938 | 0.0248 | 0.1569 | 26.705 | 18.118 | 2163 |
| Winter | h1 | 0.9433 | 0.9292 | -0.0141 | -0.0241 | 0.0005 | 12.980 | 7.220 | 2142 |
| Winter | h2 | 0.8429 | 0.8378 | -0.0051 | -0.0174 | 0.0110 | 19.646 | 11.797 | 2142 |
| Winter | h3 | 0.7341 | 0.7286 | -0.0055 | -0.0207 | 0.0101 | 25.415 | 15.436 | 2142 |
| Winter | h4 | 0.6354 | 0.6264 | -0.0090 | -0.0272 | 0.0120 | 29.821 | 18.841 | 2142 |
| Winter | h5 | 0.5367 | 0.5277 | -0.0090 | -0.0465 | 0.0256 | 33.529 | 21.589 | 2142 |
| Winter | h6 | 0.4428 | 0.4497 | 0.0069 | -0.0308 | 0.0410 | 36.192 | 23.890 | 2142 |
| Winter | h7 | 0.3949 | 0.4079 | 0.0130 | -0.0209 | 0.0422 | 37.543 | 25.179 | 2142 |
| Winter | h8 | 0.3482 | 0.3595 | 0.0113 | -0.0336 | 0.0531 | 39.045 | 26.660 | 2142 |
| Winter | h9 | 0.3435 | 0.3475 | 0.0040 | -0.0429 | 0.0495 | 39.409 | 27.009 | 2142 |
| Winter | h10 | 0.3030 | 0.3116 | 0.0086 | -0.0409 | 0.0595 | 40.478 | 27.877 | 2142 |

| Season | Horizon | Local $R^2$ | MERRA-2 $R^2$ | $\Delta R^2$ | 95% CI lower | 95% CI upper | MERRA-2 RMSE (μg m−3) | MERRA-2 MAE (μg m−3) | n |
|---|---|---|---|---|---|---|---|---|---|
| Winter | h11 | 0.2743 | 0.2426 | -0.0317 | -0.1082 | 0.0526 | 42.459 | 28.917 | 2142 |
| Winter | h12 | 0.2557 | 0.2425 | -0.0132 | -0.0573 | 0.0396 | 42.461 | 29.791 | 2142 |
| Winter | h13 | 0.1750 | 0.1476 | -0.0274 | -0.0804 | 0.0453 | 45.042 | 31.202 | 2142 |
| Winter | h14 | 0.1647 | 0.1604 | -0.0043 | -0.0502 | 0.0463 | 44.705 | 31.918 | 2142 |
| Winter | h15 | 0.1355 | 0.1255 | -0.0100 | -0.0606 | 0.0626 | 45.625 | 32.515 | 2142 |
| Winter | h16 | 0.0910 | 0.0963 | 0.0053 | -0.0686 | 0.1007 | 46.378 | 32.920 | 2142 |
| Winter | h17 | 0.0724 | 0.0832 | 0.0108 | -0.0463 | 0.0890 | 46.714 | 33.396 | 2142 |
| Winter | h18 | 0.0605 | 0.0684 | 0.0080 | -0.0403 | 0.0782 | 47.089 | 33.474 | 2142 |
| Winter | h19 | 0.0532 | 0.0547 | 0.0015 | -0.0487 | 0.0735 | 47.434 | 34.096 | 2142 |
| Winter | h20 | 0.0578 | 0.0204 | -0.0373 | -0.0946 | 0.0289 | 48.286 | 34.190 | 2142 |
| Winter | h21 | 0.0648 | 0.0483 | -0.0165 | -0.0781 | 0.0599 | 47.595 | 33.662 | 2142 |
| Winter | h22 | 0.0577 | 0.0845 | 0.0268 | -0.0325 | 0.1047 | 46.680 | 33.287 | 2142 |
| Winter | h23 | 0.0565 | 0.0705 | 0.0140 | -0.0395 | 0.0773 | 47.036 | 33.639 | 2142 |
| Winter | h24 | 0.0769 | 0.0933 | 0.0164 | -0.0544 | 0.1150 | 46.457 | 33.084 | 2142 |

*Supplementary Table S6. P90 and P95 warning performance by model and horizon*

Thresholds were estimated separately for each horizon from uncapped training targets and applied unchanged to the locked-test period.

| Model | Horizon | Rule | Training threshold (µg m−3) | Observed events | Predicted events | Precision | Recall | F1 | False alarms | Misses | n |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Local-only model | h1 | P90 | 88.8 | 213 | 203 | 0.882 | 0.840 | 0.861 | 24 | 34 | 3373 |
| Local-only model | h1 | P95 | 116.4 | 92 | 94 | 0.809 | 0.826 | 0.817 | 18 | 16 | 3373 |
| Full model (with MERRA-2) | h1 | P90 | 88.8 | 213 | 205 | 0.883 | 0.850 | 0.866 | 24 | 32 | 3373 |
| Full model (with MERRA-2) | h1 | P95 | 116.4 | 92 | 87 | 0.828 | 0.783 | 0.804 | 15 | 20 | 3373 |
| Local-only model | h2 | P90 | 88.8 | 213 | 191 | 0.780 | 0.700 | 0.738 | 42 | 64 | 3373 |
| Local-only model | h2 | P95 | 116.4 | 92 | 70 | 0.786 | 0.598 | 0.679 | 15 | 37 | 3373 |
| Full model (with MERRA-2) | h2 | P90 | 88.8 | 213 | 200 | 0.750 | 0.704 | 0.726 | 50 | 63 | 3373 |
| Full model (with MERRA-2) | h2 | P95 | 116.4 | 92 | 62 | 0.758 | 0.511 | 0.610 | 15 | 45 | 3373 |
| Local-only model | h3 | P90 | 88.8 | 213 | 175 | 0.691 | 0.568 | 0.624 | 54 | 92 | 3373 |
| Local-only model | h3 | P95 | 116.4 | 92 | 43 | 0.651 | 0.304 | 0.415 | 15 | 64 | 3373 |
| Full model (with MERRA-2) | h3 | P90 | 88.8 | 213 | 169 | 0.722 | 0.573 | 0.639 | 47 | 91 | 3373 |
| Full model (with MERRA-2) | h3 | P95 | 116.4 | 92 | 41 | 0.707 | 0.315 | 0.436 | 12 | 63 | 3373 |
| Local-only model | h4 | P90 | 88.8 | 213 | 152 | 0.632 | 0.451 | 0.526 | 56 | 117 | 3373 |
| Local-only model | h4 | P95 | 116.4 | 92 | 27 | 0.630 | 0.185 | 0.286 | 10 | 75 | 3373 |
| Full model (with MERRA-2) | h4 | P90 | 88.8 | 213 | 151 | 0.682 | 0.484 | 0.566 | 48 | 110 | 3373 |
| Full model (with MERRA-2) | h4 | P95 | 116.4 | 92 | 38 | 0.605 | 0.250 | 0.354 | 15 | 69 | 3373 |
| Local-only model | h5 | P90 | 88.8 | 213 | 142 | 0.592 | 0.394 | 0.473 | 58 | 129 | 3373 |
| Local-only model | h5 | P95 | 116.4 | 92 | 35 | 0.600 | 0.228 | 0.331 | 14 | 71 | 3373 |
| Full model (with MERRA-2) | h5 | P90 | 88.8 | 213 | 131 | 0.656 | 0.404 | 0.500 | 45 | 127 | 3373 |
| Full model (with MERRA-2) | h5 | P95 | 116.4 | 92 | 34 | 0.441 | 0.163 | 0.238 | 19 | 77 | 3373 |
| Local-only model | h6 | P90 | 88.8 | 213 | 100 | 0.580 | 0.272 | 0.371 | 42 | 155 | 3373 |
| Local-only model | h6 | P95 | 116.4 | 92 | 19 | 0.526 | 0.109 | 0.180 | 9 | 82 | 3373 |
| Full model (with MERRA-2) | h6 | P90 | 88.8 | 213 | 80 | 0.588 | 0.221 | 0.321 | 33 | 166 | 3373 |
| Full model (with MERRA-2) | h6 | P95 | 116.4 | 92 | 13 | 0.385 | 0.054 | 0.095 | 8 | 87 | 3373 |
| Local-only model | h7 | P90 | 88.8 | 213 | 64 | 0.516 | 0.155 | 0.238 | 31 | 180 | 3373 |
| Local-only model | h7 | P95 | 116.4 | 92 | 5 | 0.200 | 0.011 | 0.021 | 4 | 91 | 3373 |
| Full model (with MERRA-2) | h7 | P90 | 88.8 | 213 | 108 | 0.519 | 0.263 | 0.349 | 52 | 157 | 3373 |
| Full model (with MERRA-2) | h7 | P95 | 116.4 | 92 | 25 | 0.240 | 0.065 | 0.103 | 19 | 86 | 3373 |
| Local-only model | h8 | P90 | 88.8 | 213 | 70 | 0.514 | 0.169 | 0.254 | 34 | 177 | 3373 |
| Local-only model | h8 | P95 | 116.4 | 92 | 5 | 0.000 | 0.000 | 0.000 | 5 | 92 | 3373 |
| Full model (with MERRA-2) | h8 | P90 | 88.8 | 213 | 80 | 0.550 | 0.207 | 0.300 | 36 | 169 | 3373 |
| Full model (with MERRA-2) | h8 | P95 | 116.4 | 92 | 23 | 0.174 | 0.043 | 0.070 | 19 | 88 | 3373 |
| Local-only model | h9 | P90 | 88.8 | 213 | 33 | 0.515 | 0.080 | 0.138 | 16 | 196 | 3373 |
| Local-only model | h9 | P95 | 116.4 | 92 | 6 | 0.000 | 0.000 | 0.000 | 6 | 92 | 3373 |
| Full model (with MERRA-2) | h9 | P90 | 88.8 | 213 | 96 | 0.500 | 0.225 | 0.311 | 48 | 165 | 3373 |
| Full model (with MERRA-2) | h9 | P95 | 116.4 | 92 | 18 | 0.222 | 0.043 | 0.073 | 14 | 88 | 3373 |
| Local-only model | h10 | P90 | 88.8 | 213 | 30 | 0.567 | 0.080 | 0.140 | 13 | 196 | 3373 |
| Local-only model | h10 | P95 | 116.4 | 92 | 1 | 0.000 | 0.000 | 0.000 | 1 | 92 | 3373 |
| Full model (with MERRA-2) | h10 | P90 | 88.8 | 213 | 86 | 0.500 | 0.202 | 0.288 | 43 | 170 | 3373 |
| Full model (with MERRA-2) | h10 | P95 | 116.4 | 92 | 14 | 0.071 | 0.011 | 0.019 | 13 | 91 | 3373 |

| Model | Horizon | Rule | Training threshold (µg m−3) | Observed events | Predicted events | Precision | Recall | F1 | False alarms | Misses | n |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Local-only model | h11 | P90 | 88.8 | 213 | 44 | 0.341 | 0.070 | 0.117 | 29 | 198 | 3373 |
| Local-only model | h11 | P95 | 116.4 | 90 | 2 | 0.000 | 0.000 | 0.000 | 2 | 90 | 3373 |
| Full model (with MERRA-2) | h11 | P90 | 88.8 | 213 | 87 | 0.414 | 0.169 | 0.240 | 51 | 177 | 3373 |
| Full model (with MERRA-2) | h11 | P95 | 116.4 | 90 | 1 | 0.000 | 0.000 | 0.000 | 1 | 90 | 3373 |
| Local-only model | h12 | P90 | 88.8 | 213 | 16 | 0.250 | 0.019 | 0.035 | 12 | 209 | 3373 |
| Local-only model | h12 | P95 | 116.4 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Full model (with MERRA-2) | h12 | P90 | 88.8 | 213 | 48 | 0.438 | 0.099 | 0.161 | 27 | 192 | 3373 |
| Full model (with MERRA-2) | h12 | P95 | 116.4 | 90 | 2 | 0.000 | 0.000 | 0.000 | 2 | 90 | 3373 |
| Local-only model | h13 | P90 | 88.8 | 213 | 21 | 0.524 | 0.052 | 0.094 | 10 | 202 | 3373 |
| Local-only model | h13 | P95 | 116.5 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Full model (with MERRA-2) | h13 | P90 | 88.8 | 213 | 84 | 0.429 | 0.169 | 0.242 | 48 | 177 | 3373 |
| Full model (with MERRA-2) | h13 | P95 | 116.5 | 90 | 7 | 0.286 | 0.022 | 0.041 | 5 | 88 | 3373 |
| Local-only model | h14 | P90 | 88.8 | 213 | 8 | 0.250 | 0.009 | 0.018 | 6 | 211 | 3373 |
| Local-only model | h14 | P95 | 116.5 | 90 | 1 | 0.000 | 0.000 | 0.000 | 1 | 90 | 3373 |
| Full model (with MERRA-2) | h14 | P90 | 88.8 | 213 | 47 | 0.489 | 0.108 | 0.177 | 24 | 190 | 3373 |
| Full model (with MERRA-2) | h14 | P95 | 116.5 | 90 | 7 | 0.000 | 0.000 | 0.000 | 7 | 90 | 3373 |
| Local-only model | h15 | P90 | 88.8 | 213 | 6 | 0.167 | 0.005 | 0.009 | 5 | 212 | 3373 |
| Local-only model | h15 | P95 | 116.5 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Full model (with MERRA-2) | h15 | P90 | 88.8 | 213 | 62 | 0.355 | 0.103 | 0.160 | 40 | 191 | 3373 |
| Full model (with MERRA-2) | h15 | P95 | 116.5 | 90 | 4 | 0.000 | 0.000 | 0.000 | 4 | 90 | 3373 |
| Local-only model | h16 | P90 | 88.8 | 213 | 1 | 0.000 | 0.000 | 0.000 | 1 | 213 | 3373 |
| Local-only model | h16 | P95 | 116.5 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Full model (with MERRA-2) | h16 | P90 | 88.8 | 213 | 30 | 0.633 | 0.089 | 0.156 | 11 | 194 | 3373 |
| Full model (with MERRA-2) | h16 | P95 | 116.5 | 90 | 2 | 0.000 | 0.000 | 0.000 | 2 | 90 | 3373 |
| Local-only model | h17 | P90 | 88.8 | 213 | 3 | 0.667 | 0.009 | 0.019 | 1 | 211 | 3373 |
| Local-only model | h17 | P95 | 116.5 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Full model (with MERRA-2) | h17 | P90 | 88.8 | 213 | 16 | 0.625 | 0.047 | 0.087 | 6 | 203 | 3373 |
| Full model (with MERRA-2) | h17 | P95 | 116.5 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Local-only model | h18 | P90 | 88.8 | 213 | 1 | 1.000 | 0.005 | 0.009 | 0 | 212 | 3373 |
| Local-only model | h18 | P95 | 116.6 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Full model (with MERRA-2) | h18 | P90 | 88.8 | 213 | 16 | 0.812 | 0.061 | 0.114 | 3 | 200 | 3373 |
| Full model (with MERRA-2) | h18 | P95 | 116.6 | 90 | 3 | 0.000 | 0.000 | 0.000 | 3 | 90 | 3373 |
| Local-only model | h19 | P90 | 88.8 | 213 | 3 | 0.333 | 0.005 | 0.009 | 2 | 212 | 3373 |
| Local-only model | h19 | P95 | 116.6 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Full model (with MERRA-2) | h19 | P90 | 88.8 | 213 | 31 | 0.613 | 0.089 | 0.156 | 12 | 194 | 3373 |
| Full model (with MERRA-2) | h19 | P95 | 116.6 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Local-only model | h20 | P90 | 88.8 | 213 | 9 | 0.222 | 0.009 | 0.018 | 7 | 211 | 3373 |
| Local-only model | h20 | P95 | 116.6 | 90 | 2 | 0.000 | 0.000 | 0.000 | 2 | 90 | 3373 |
| Full model (with MERRA-2) | h20 | P90 | 88.8 | 213 | 35 | 0.657 | 0.108 | 0.185 | 12 | 190 | 3373 |
| Full model (with MERRA-2) | h20 | P95 | 116.6 | 90 | 1 | 0.000 | 0.000 | 0.000 | 1 | 90 | 3373 |
| Local-only model | h21 | P90 | 88.6 | 213 | 4 | 0.250 | 0.005 | 0.009 | 3 | 212 | 3373 |
| Local-only model | h21 | P95 | 116.6 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Full model (with MERRA-2) | h21 | P90 | 88.6 | 213 | 38 | 0.421 | 0.075 | 0.127 | 22 | 197 | 3373 |
| Full model (with MERRA-2) | h21 | P95 | 116.6 | 90 | 6 | 0.000 | 0.000 | 0.000 | 6 | 90 | 3373 |
| Local-only model | h22 | P90 | 88.6 | 213 | 3 | 0.667 | 0.009 | 0.019 | 1 | 211 | 3373 |

| Model | Horizon | Rule | Training threshold (µg m−3) | Observed events | Predicted events | Precision | Recall | F1 | False alarms | Misses | n |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Local-only model | h22 | P95 | 116.6 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Full model (with MERRA-2) | h22 | P90 | 88.6 | 213 | 33 | 0.515 | 0.080 | 0.138 | 16 | 196 | 3373 |
| Full model (with MERRA-2) | h22 | P95 | 116.6 | 90 | 7 | 0.143 | 0.011 | 0.021 | 6 | 89 | 3373 |
| Local-only model | h23 | P90 | 88.7 | 213 | 10 | 0.100 | 0.005 | 0.009 | 9 | 212 | 3373 |
| Local-only model | h23 | P95 | 116.7 | 90 | 6 | 0.000 | 0.000 | 0.000 | 6 | 90 | 3373 |
| Full model (with MERRA-2) | h23 | P90 | 88.7 | 213 | 22 | 0.500 | 0.052 | 0.094 | 11 | 202 | 3373 |
| Full model (with MERRA-2) | h23 | P95 | 116.7 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Local-only model | h24 | P90 | 88.6 | 214 | 4 | 0.250 | 0.005 | 0.009 | 3 | 213 | 3373 |
| Local-only model | h24 | P95 | 116.7 | 90 | 0 | 0.000 | 0.000 | 0.000 | 0 | 90 | 3373 |
| Full model (with MERRA-2) | h24 | P90 | 88.6 | 214 | 37 | 0.459 | 0.079 | 0.135 | 20 | 197 | 3373 |
| Full model (with MERRA-2) | h24 | P95 | 116.7 | 90 | 1 | 0.000 | 0.000 | 0.000 | 1 | 90 | 3373 |

*Supplementary Table S7A. Locked-test ablation utility and exact-SHAP family shares*

Ablation values are full-model $R^2$ minus $R^2$ after removal. The detailed ablation values in this table are retained from the frozen Draft 2 secondary package; they are supporting evidence and did not select the Draft 3 primary model. SHAP columns are percentages of total mean absolute attribution at each horizon.

| Horizon | MERRA-2 ablation $\Delta R^2$ | Engineered local-condition indicator ablation $\Delta R^2$ | Calendar/seasonal (%) | Engineered local-condition indicators (%) | Target calendar (%) | Local meteorology (%) | MERRA-2 (%) | Other (%) | Other pollutants (%) | PM2.5 history (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| h1 | -0.0022 | 0.0045 | 10.24 | 7.88 | 1.18 | 11.73 | 5.35 | 0.00 | 10.25 | 53.37 |
| h2 | -0.0037 | 0.0133 | 13.16 | 12.63 | 3.68 | 12.38 | 10.59 | 0.00 | 14.82 | 32.75 |
| h3 | 0.0025 | 0.0266 | 15.10 | 14.38 | 4.03 | 12.29 | 14.28 | 0.00 | 13.55 | 26.36 |
| h4 | 0.0173 | 0.0413 | 17.13 | 12.47 | 3.60 | 14.53 | 16.25 | 0.00 | 15.82 | 20.19 |
| h5 | 0.0032 | 0.0346 | 13.87 | 12.01 | 6.46 | 13.76 | 19.55 | 0.00 | 15.71 | 18.64 |
| h6 | 0.0217 | 0.0354 | 14.47 | 13.24 | 5.48 | 13.93 | 24.19 | 0.00 | 18.42 | 10.29 |
| h7 | 0.0232 | 0.0320 | 13.84 | 8.90 | 7.87 | 18.26 | 22.72 | 0.00 | 14.52 | 13.89 |
| h8 | 0.0305 | 0.0088 | 17.63 | 7.65 | 3.55 | 18.76 | 25.12 | 0.00 | 14.73 | 12.56 |
| h9 | 0.0405 | 0.0452 | 16.94 | 5.43 | 4.47 | 19.74 | 24.19 | 0.00 | 19.19 | 10.05 |
| h10 | 0.0584 | 0.0083 | 15.78 | 5.19 | 4.79 | 19.44 | 25.31 | 0.00 | 20.36 | 9.12 |
| h11 | 0.0620 | 0.0117 | 17.60 | 3.58 | 5.33 | 19.92 | 29.36 | 0.00 | 15.82 | 8.39 |
| h12 | 0.0561 | -0.0041 | 16.64 | 3.73 | 5.99 | 20.39 | 26.62 | 0.41 | 17.54 | 8.68 |
| h13 | 0.0870 | 0.0183 | 18.36 | 3.55 | 4.60 | 18.85 | 28.43 | 0.00 | 16.73 | 9.47 |
| h14 | 0.0905 | 0.0130 | 19.52 | 3.96 | 4.07 | 18.71 | 31.49 | 0.10 | 15.75 | 6.42 |
| h15 | 0.0917 | -0.0182 | 20.01 | 4.31 | 2.96 | 18.38 | 32.39 | 0.08 | 16.19 | 5.68 |
| h16 | 0.0906 | -0.0038 | 18.12 | 4.40 | 4.12 | 18.79 | 31.24 | 0.00 | 17.78 | 5.56 |
| h17 | 0.0691 | -0.0168 | 14.20 | 4.24 | 6.74 | 19.33 | 29.81 | 0.00 | 18.83 | 6.85 |
| h18 | 0.0911 | -0.0475 | 14.17 | 3.41 | 6.96 | 21.86 | 29.12 | 0.00 | 17.69 | 6.79 |
| h19 | 0.1296 | 0.0109 | 13.72 | 3.13 | 5.40 | 22.39 | 28.67 | 0.00 | 19.02 | 7.67 |
| h20 | 0.0686 | -0.0030 | 13.37 | 3.42 | 6.31 | 20.93 | 33.79 | 0.00 | 16.44 | 5.75 |
| h21 | 0.0903 | 0.0236 | 17.68 | 3.45 | 5.67 | 20.64 | 31.79 | 0.00 | 13.41 | 7.35 |
| h22 | -0.0013 | -0.0038 | 13.76 | 3.76 | 6.15 | 21.73 | 32.37 | 0.00 | 14.76 | 7.47 |
| h23 | 0.0709 | 0.0406 | 15.25 | 6.53 | 5.34 | 20.70 | 30.47 | 0.00 | 15.01 | 6.71 |
| h24 | 0.0248 | -0.0265 | 12.29 | 5.55 | 6.70 | 23.05 | 32.31 | 0.00 | 14.89 | 5.20 |

*Supplementary Table S7B. Seasonal MERRA-2 attribution and ablation utility*

Seasonal values use fixed-parameter rolling-origin models. Winter values use the coverage-corrected composite.

| Season | Horizon | MERRA-2 SHAP share (%) | MERRA-2 ablation $\Delta R^2$ | Engineered local-condition indicator ablation $\Delta R^2$ |
|---|---|---|---|---|
| Autumn | h1 | 4.20 | -0.0020 | 0.0016 |
| Autumn | h2 | 7.51 | 0.0017 | 0.0185 |
| Autumn | h3 | 11.46 | -0.0034 | 0.0230 |
| Autumn | h4 | 14.05 | 0.0071 | 0.0400 |
| Autumn | h5 | 15.55 | 0.0014 | 0.0452 |
| Autumn | h6 | 18.12 | 0.0066 | 0.0544 |
| Autumn | h7 | 18.87 | -0.0070 | 0.0352 |
| Autumn | h8 | 21.76 | 0.0042 | 0.0140 |
| Autumn | h9 | 22.66 | 0.0308 | 0.0199 |
| Autumn | h10 | 23.39 | 0.0523 | 0.0092 |
| Autumn | h11 | 24.87 | 0.0518 | 0.0096 |
| Autumn | h12 | 26.29 | 0.0242 | 0.0142 |
| Autumn | h13 | 26.40 | 0.0414 | 0.0040 |
| Autumn | h14 | 27.04 | 0.0744 | 0.0184 |
| Autumn | h15 | 27.63 | 0.0866 | -0.0029 |
| Autumn | h16 | 27.82 | 0.0930 | 0.0170 |
| Autumn | h17 | 28.84 | 0.1118 | -0.0021 |
| Autumn | h18 | 29.53 | 0.1098 | -0.0165 |
| Autumn | h19 | 28.12 | 0.0983 | -0.0116 |
| Autumn | h20 | 28.98 | 0.0738 | 0.0019 |
| Autumn | h21 | 30.54 | 0.0610 | 0.0001 |
| Autumn | h22 | 30.38 | 0.0799 | 0.0179 |
| Autumn | h23 | 29.88 | 0.0639 | 0.0038 |
| Autumn | h24 | 29.50 | 0.0938 | 0.0288 |
| Spring | h1 | 5.56 | -0.0008 | 0.0037 |
| Spring | h2 | 10.01 | -0.0031 | 0.0156 |
| Spring | h3 | 13.66 | 0.0071 | 0.0187 |
| Spring | h4 | 17.00 | 0.0063 | 0.0181 |
| Spring | h5 | 20.95 | 0.0053 | 0.0284 |
| Spring | h6 | 23.44 | -0.0043 | 0.0204 |
| Spring | h7 | 24.57 | 0.0166 | 0.0167 |
| Spring | h8 | 25.27 | -0.0083 | 0.0142 |
| Spring | h9 | 28.40 | -0.0018 | 0.0177 |
| Spring | h10 | 30.61 | -0.0217 | -0.0233 |
| Spring | h11 | 30.53 | -0.0130 | -0.0053 |
| Spring | h12 | 30.43 | -0.0126 | 0.0032 |
| Spring | h13 | 31.73 | -0.0092 | -0.0064 |
| Spring | h14 | 33.50 | -0.0314 | 0.0256 |
| Spring | h15 | 33.92 | 0.0086 | 0.0215 |
| Spring | h16 | 34.60 | -0.0105 | -0.0120 |

| Season | Horizon | MERRA-2 SHAP share (%) | MERRA-2 ablation ΔR² | Engineered local-condition indicator ablation ΔR² |
|---|---|---|---|---|
| Spring | h17 | 32.38 | -0.0066 | 0.0216 |
| Spring | h18 | 32.36 | -0.0063 | 0.0173 |
| Spring | h19 | 33.70 | -0.0071 | -0.0287 |
| Spring | h20 | 32.27 | 0.0352 | -0.0037 |
| Spring | h21 | 34.30 | 0.0402 | 0.0078 |
| Spring | h22 | 34.02 | 0.0368 | -0.0146 |
| Spring | h23 | 29.77 | 0.0314 | 0.0080 |
| Spring | h24 | 30.32 | 0.0516 | 0.0156 |
| Summer | h1 | 3.84 | 0.0007 | -0.0008 |
| Summer | h2 | 8.38 | -0.0008 | -0.0003 |
| Summer | h3 | 10.99 | 0.0052 | 0.0046 |
| Summer | h4 | 12.84 | -0.0029 | -0.0004 |
| Summer | h5 | 14.84 | 0.0068 | -0.0083 |
| Summer | h6 | 17.65 | 0.0028 | 0.0043 |
| Summer | h7 | 19.59 | 0.0055 | -0.0046 |
| Summer | h8 | 20.54 | 0.0003 | 0.0015 |
| Summer | h9 | 22.42 | 0.0167 | 0.0025 |
| Summer | h10 | 23.75 | 0.0157 | -0.0030 |
| Summer | h11 | 21.57 | 0.0133 | -0.0216 |
| Summer | h12 | 21.98 | 0.0575 | 0.0139 |
| Summer | h13 | 23.59 | 0.0407 | -0.0089 |
| Summer | h14 | 24.79 | 0.0418 | -0.0042 |
| Summer | h15 | 24.31 | 0.0172 | 0.0142 |
| Summer | h16 | 25.07 | 0.0324 | -0.0017 |
| Summer | h17 | 26.54 | 0.0405 | -0.0287 |
| Summer | h18 | 26.34 | 0.0500 | 0.0073 |
| Summer | h19 | 25.00 | 0.0232 | -0.0029 |
| Summer | h20 | 26.46 | 0.0320 | 0.0185 |
| Summer | h21 | 29.27 | 0.0274 | -0.0087 |
| Summer | h22 | 24.68 | 0.0339 | 0.0103 |
| Summer | h23 | 25.71 | 0.0232 | 0.0103 |
| Summer | h24 | 25.31 | 0.0034 | -0.0239 |
| Winter | h1 | 5.88 | -0.0141 | -0.0002 |
| Winter | h2 | 10.41 | -0.0051 | 0.0019 |
| Winter | h3 | 13.74 | -0.0055 | 0.0083 |
| Winter | h4 | 16.70 | -0.0090 | 0.0196 |
| Winter | h5 | 19.11 | -0.0090 | 0.0041 |
| Winter | h6 | 21.44 | 0.0069 | 0.0082 |
| Winter | h7 | 23.20 | 0.0130 | 0.0059 |
| Winter | h8 | 24.09 | 0.0113 | -0.0007 |
| Winter | h9 | 26.65 | 0.0040 | 0.0165 |
| Winter | h10 | 26.54 | 0.0086 | -0.0051 |
| Winter | h11 | 26.92 | -0.0317 | -0.0369 |
| Winter | h12 | 27.88 | -0.0132 | 0.0303 |
| Winter | h13 | 27.71 | -0.0274 | -0.0155 |

| Season | Horizon | MERRA-2 SHAP share (%) | MERRA-2 ablation ΔR² | Engineered local-condition indicator ablation ΔR² |
|---|---|---|---|---|
| Winter | h14 | 28.71 | -0.0043 | 0.0056 |
| Winter | h15 | 28.51 | -0.0100 | 0.0081 |
| Winter | h16 | 27.71 | 0.0053 | -0.0275 |
| Winter | h17 | 29.17 | 0.0108 | 0.0213 |
| Winter | h18 | 27.51 | 0.0080 | -0.0107 |
| Winter | h19 | 28.30 | 0.0015 | -0.0223 |
| Winter | h20 | 28.66 | -0.0373 | -0.0217 |
| Winter | h21 | 29.00 | -0.0165 | -0.0145 |
| Winter | h22 | 28.71 | 0.0268 | 0.0263 |
| Winter | h23 | 28.26 | 0.0140 | -0.0202 |
| Winter | h24 | 29.08 | 0.0164 | 0.0021 |

*Supplementary Table S8. Frozen MERRA-2 spatial and temporal alignment sensitivities*

All scenarios reused the primary full-model hyperparameters. Scenario roles were frozen before evaluation, keeping the primary model unchanged.

| Scenario | Mean $R^2$ | Mean RMSE (µg m−3) | Mean MAE (µg m−3) | Mean bias (µg m−3) | $\Delta R^2$ vs primary | ΔRMSE vs primary | ΔMAE vs primary |
|---|---|---|---|---|---|---|---|
| primary availability-aligned area weighted cst hour end | 0.4563 | 22.323 | 15.061 | 1.846 | +0.0000 | +0.0000 | +0.0000 |
| spatial sensitivity nearest cell cst hour end | 0.4518 | 22.402 | 15.113 | 1.867 | -0.0045 | +0.0782 | +0.0521 |
| retrospective timing sensitivity area weighted cst hour start | 0.4582 | 22.291 | 15.055 | 2.005 | +0.0018 | -0.0324 | -0.0056 |
| retrospective combined sensitivity nearest cell cst hour start | 0.4558 | 22.339 | 15.048 | 1.906 | -0.0006 | +0.0159 | -0.0130 |

*Supplementary Table S9A. Prospectively specified feature-reduction acceptance audit*

The prospectively specified replacement required all three criteria to pass at every horizon. Seventeen of 24 horizons passed jointly, so the full 222-predictor model remained primary.

| Horizon | Band | Reduced predictors | ΔR² | Relative RMSE change | Relative MAE change | R² pass | RMSE pass | MAE pass | Joint pass |
|---|---|---|---|---|---|---|---|---|---|
| h1 | h1–h6 | 119 | -0.0024 | 0.0112 | 0.0166 | Yes | No | No | No |
| h2 | h1–h6 | 119 | 0.0073 | -0.0148 | -0.0086 | Yes | Yes | Yes | Yes |
| h3 | h1–h6 | 119 | 0.0049 | -0.0068 | -0.0106 | Yes | Yes | Yes | Yes |
| h4 | h1–h6 | 119 | 0.0092 | -0.0101 | -0.0048 | Yes | Yes | Yes | Yes |
| h5 | h1–h6 | 119 | 0.0134 | -0.0128 | 0.0036 | Yes | Yes | Yes | Yes |
| h6 | h1–h6 | 119 | 0.0001 | -0.0001 | 0.0031 | Yes | Yes | Yes | Yes |
| h7 | h7–h12 | 130 | -0.0099 | 0.0082 | -0.0022 | Yes | Yes | Yes | Yes |
| h8 | h7–h12 | 130 | -0.0120 | 0.0095 | 0.0147 | No | Yes | No | No |
| h9 | h7–h12 | 130 | 0.0127 | -0.0096 | 0.0047 | Yes | Yes | Yes | Yes |
| h10 | h7–h12 | 130 | 0.0096 | -0.0071 | -0.0163 | Yes | Yes | Yes | Yes |
| h11 | h7–h12 | 130 | 0.0008 | -0.0006 | -0.0139 | Yes | Yes | Yes | Yes |
| h12 | h7–h12 | 130 | -0.0092 | 0.0066 | -0.0074 | Yes | Yes | Yes | Yes |
| h13 | h13–h24 | 140 | 0.0017 | -0.0012 | 0.0074 | Yes | Yes | Yes | Yes |
| h14 | h13–h24 | 140 | -0.0067 | 0.0046 | 0.0069 | Yes | Yes | Yes | Yes |
| h15 | h13–h24 | 140 | -0.0001 | 0.0001 | 0.0098 | Yes | Yes | Yes | Yes |
| h16 | h13–h24 | 140 | -0.0035 | 0.0023 | 0.0114 | Yes | Yes | No | No |
| h17 | h13–h24 | 140 | -0.0234 | 0.0151 | 0.0207 | No | No | No | No |
| h18 | h13–h24 | 140 | -0.0248 | 0.0159 | 0.0081 | No | No | Yes | No |
| h19 | h13–h24 | 140 | 0.0022 | -0.0014 | -0.0022 | Yes | Yes | Yes | Yes |
| h20 | h13–h24 | 140 | -0.0049 | 0.0031 | 0.0012 | Yes | Yes | Yes | Yes |
| h21 | h13–h24 | 140 | 0.0004 | -0.0003 | -0.0066 | Yes | Yes | Yes | Yes |
| h22 | h13–h24 | 140 | 0.0076 | -0.0048 | -0.0093 | Yes | Yes | Yes | Yes |
| h23 | h13–h24 | 140 | -0.0191 | 0.0123 | 0.0110 | No | No | No | No |
| h24 | h13–h24 | 140 | -0.0193 | 0.0123 | 0.0094 | No | No | Yes | No |

*Supplementary Table S9B. Post-hoc 59-predictor validation profile*

The 59-predictor budget remains validation-only and requires new-period testing. For $\Delta R^2$, negative values mean the reduced model was below the full model.

| Horizon | Predictors | Reduced model minus full model $\Delta R^2$ | Relative RMSE change | Relative MAE change | Post-hoc 5% gate |
|---|---|---|---|---|---|
| h1 | 59 | -0.0041 | 0.0188 | 0.0143 | Yes |
| h2 | 59 | -0.0040 | 0.0081 | -0.0029 | Yes |
| h3 | 59 | -0.0088 | 0.0121 | 0.0083 | Yes |
| h4 | 59 | -0.0055 | 0.0061 | 0.0026 | Yes |
| h5 | 59 | 0.0134 | -0.0128 | -0.0062 | Yes |
| h6 | 59 | -0.0118 | 0.0102 | 0.0052 | Yes |
| h7 | 59 | -0.0076 | 0.0062 | -0.0062 | Yes |
| h8 | 59 | 0.0069 | -0.0054 | -0.0064 | Yes |
| h9 | 59 | -0.0037 | 0.0028 | 0.0022 | Yes |
| h10 | 59 | -0.0229 | 0.0171 | 0.0388 | Yes |
| h11 | 59 | -0.0086 | 0.0063 | 0.0194 | Yes |
| h12 | 59 | -0.0124 | 0.0088 | 0.0179 | Yes |
| h13 | 59 | -0.0201 | 0.0142 | 0.0345 | Yes |
| h14 | 59 | -0.0120 | 0.0083 | 0.0263 | Yes |
| h15 | 59 | 0.0033 | -0.0022 | 0.0164 | Yes |
| h16 | 59 | -0.0024 | 0.0015 | 0.0154 | Yes |
| h17 | 59 | 0.0034 | -0.0022 | 0.0156 | Yes |
| h18 | 59 | 0.0019 | -0.0012 | 0.0213 | Yes |
| h19 | 59 | 0.0032 | -0.0020 | 0.0128 | Yes |
| h20 | 59 | 0.0053 | -0.0034 | -0.0013 | Yes |
| h21 | 59 | -0.0056 | 0.0035 | 0.0284 | Yes |
| h22 | 59 | 0.0029 | -0.0018 | 0.0019 | Yes |
| h23 | 59 | -0.0004 | 0.0003 | 0.0091 | Yes |
| h24 | 59 | -0.0237 | 0.0151 | 0.0302 | Yes |

*Supplementary Table S9C. Post-hoc 59-predictor stress-test intervals and pass frequencies*

Paired 28 h moving-block bootstrap intervals and 30 deterministic-seed refits were evaluated at h10, h13, and h24.

| Horizon | Metric | Reduced model minus full model difference | 95% CI lower | 95% CI upper | CI excludes zero | Bootstrap joint-pass frequency | Seed joint-pass frequency | Seeds passing (of 30) |
|---|---|---|---|---|---|---|---|---|
| h10 | MAE | 0.3880 | 0.1767 | 0.5806 | Yes | 0.833 | 0.933 | 28 |
| h10 | R2 | -0.0238 | -0.0549 | -0.0010 | Yes | 0.833 | 0.933 | 28 |
| h10 | RMSE | 0.2749 | 0.0125 | 0.5716 | Yes | 0.833 | 0.933 | 28 |
| h13 | MAE | 0.3455 | 0.1064 | 0.5891 | Yes | 0.901 | 1.000 | 30 |
| h13 | R2 | -0.0209 | -0.0521 | 0.0005 | No | 0.901 | 1.000 | 30 |
| h13 | RMSE | 0.2300 | -0.0076 | 0.5072 | No | 0.901 | 1.000 | 30 |
| h24 | MAE | 0.3392 | -0.0090 | 0.6673 | No | 0.893 | 0.967 | 29 |
| h24 | R2 | -0.0238 | -0.0561 | 0.0059 | No | 0.893 | 0.967 | 29 |
| h24 | RMSE | 0.2578 | -0.0647 | 0.5543 | No | 0.893 | 0.967 | 29 |

*Supplementary Table S9D. Feature-reduction robustness-tier counts*

Counts summarize the complete 666 predictor-band records (222 predictors across three horizon bands) from the prospectively specified stability analysis.

| Band | Feature family | Robustness tier | Predictor-band records |
|---|---|---|---|
| h13–h24 | Calendar/seasonal timing | core_stable_both | 9 |
| h13–h24 | Calendar/seasonal timing | correlated_redundant_candidate | 8 |
| h13–h24 | Calendar/seasonal timing | low_evidence_nonredundant_candidate | 2 |
| h13–h24 | Calendar/seasonal timing | supporting_shap_or_stable_permutation | 10 |
| h13–h24 | Engineered local-condition indicators | core_stable_both | 1 |
| h13–h24 | Engineered local-condition indicators | correlated_redundant_candidate | 2 |
| h13–h24 | Engineered local-condition indicators | low_evidence_nonredundant_candidate | 18 |
| h13–h24 | Engineered local-condition indicators | supporting_shap_or_stable_permutation | 2 |
| h13–h24 | Known target-calendar timing | core_stable_both | 3 |
| h13–h24 | Known target-calendar timing | low_evidence_nonredundant_candidate | 4 |
| h13–h24 | Known target-calendar timing | supporting_shap_or_stable_permutation | 2 |
| h13–h24 | Local meteorology | core_stable_both | 4 |
| h13–h24 | Local meteorology | correlated_redundant_candidate | 5 |
| h13–h24 | Local meteorology | low_evidence_nonredundant_candidate | 12 |
| h13–h24 | Local meteorology | supporting_shap_or_stable_permutation | 15 |
| h13–h24 | MERRA-2 aerosol/reanalysis | core_stable_both | 17 |
| h13–h24 | MERRA-2 aerosol/reanalysis | correlated_redundant_candidate | 11 |
| h13–h24 | MERRA-2 aerosol/reanalysis | low_evidence_nonredundant_candidate | 1 |
| h13–h24 | MERRA-2 aerosol/reanalysis | supporting_shap_or_stable_permutation | 44 |
| h13–h24 | Other | correlated_redundant_candidate | 1 |
| h13–h24 | Other pollutant history | core_stable_both | 13 |
| h13–h24 | Other pollutant history | correlated_redundant_candidate | 3 |
| h13–h24 | Other pollutant history | low_evidence_nonredundant_candidate | 11 |
| h13–h24 | Other pollutant history | supporting_shap_or_stable_permutation | 10 |
| h13–h24 | PM2.5 history | core_stable_both | 8 |
| h13–h24 | PM2.5 history | low_evidence_nonredundant_candidate | 4 |
| h13–h24 | PM2.5 history | supporting_shap_or_stable_permutation | 2 |
| h7–h12 | Calendar/seasonal timing | core_stable_both | 8 |
| h7–h12 | Calendar/seasonal timing | correlated_redundant_candidate | 6 |
| h7–h12 | Calendar/seasonal timing | low_evidence_nonredundant_candidate | 5 |
| h7–h12 | Calendar/seasonal timing | supporting_shap_or_stable_permutation | 10 |
| h7–h12 | Engineered local-condition indicators | core_stable_both | 2 |
| h7–h12 | Engineered local-condition indicators | correlated_redundant_candidate | 1 |
| h7–h12 | Engineered local-condition indicators | low_evidence_nonredundant_candidate | 16 |
| h7–h12 | Engineered local-condition indicators | supporting_shap_or_stable_permutation | 4 |
| h7–h12 | Known target-calendar timing | core_stable_both | 2 |
| h7–h12 | Known target-calendar timing | low_evidence_nonredundant_candidate | 4 |
| h7–h12 | Known target-calendar timing | supporting_shap_or_stable_permutation | 3 |
| h7–h12 | Local meteorology | core_stable_both | 6 |
| h7–h12 | Local meteorology | correlated_redundant_candidate | 4 |

| Band | Feature family | Robustness tier | Predictor-band records |
|---|---|---|---|
| h7–h12 | Local meteorology | low_evidence_nonredundant_candidate | 11 |
| h7–h12 | Local meteorology | supporting_shap_or_stable_permutation | 15 |
| h7–h12 | MERRA-2 aerosol/reanalysis | core_stable_both | 16 |
| h7–h12 | MERRA-2 aerosol/reanalysis | correlated_redundant_candidate | 17 |
| h7–h12 | MERRA-2 aerosol/reanalysis | low_evidence_nonredundant_candidate | 3 |
| h7–h12 | MERRA-2 aerosol/reanalysis | supporting_shap_or_stable_permutation | 37 |
| h7–h12 | Other | correlated_redundant_candidate | 1 |
| h7–h12 | Other pollutant history | core_stable_both | 12 |
| h7–h12 | Other pollutant history | correlated_redundant_candidate | 3 |
| h7–h12 | Other pollutant history | low_evidence_nonredundant_candidate | 15 |
| h7–h12 | Other pollutant history | supporting_shap_or_stable_permutation | 7 |
| h7–h12 | PM2.5 history | core_stable_both | 5 |
| h7–h12 | PM2.5 history | correlated_redundant_candidate | 1 |
| h7–h12 | PM2.5 history | low_evidence_nonredundant_candidate | 5 |
| h7–h12 | PM2.5 history | supporting_shap_or_stable_permutation | 3 |
| h1–h6 | Calendar/seasonal timing | core_stable_both | 7 |
| h1–h6 | Calendar/seasonal timing | correlated_redundant_candidate | 4 |
| h1–h6 | Calendar/seasonal timing | low_evidence_nonredundant_candidate | 6 |
| h1–h6 | Calendar/seasonal timing | supporting_shap_or_stable_permutation | 12 |
| h1–h6 | Engineered local-condition indicators | core_stable_both | 5 |
| h1–h6 | Engineered local-condition indicators | correlated_redundant_candidate | 1 |
| h1–h6 | Engineered local-condition indicators | low_evidence_nonredundant_candidate | 14 |
| h1–h6 | Engineered local-condition indicators | supporting_shap_or_stable_permutation | 3 |
| h1–h6 | Known target-calendar timing | core_stable_both | 2 |
| h1–h6 | Known target-calendar timing | low_evidence_nonredundant_candidate | 4 |
| h1–h6 | Known target-calendar timing | supporting_shap_or_stable_permutation | 3 |
| h1–h6 | Local meteorology | core_stable_both | 4 |
| h1–h6 | Local meteorology | correlated_redundant_candidate | 4 |
| h1–h6 | Local meteorology | low_evidence_nonredundant_candidate | 13 |
| h1–h6 | Local meteorology | supporting_shap_or_stable_permutation | 15 |
| h1–h6 | MERRA-2 aerosol/reanalysis | correlated_redundant_candidate | 23 |
| h1–h6 | MERRA-2 aerosol/reanalysis | low_evidence_nonredundant_candidate | 13 |
| h1–h6 | MERRA-2 aerosol/reanalysis | supporting_shap_or_stable_permutation | 37 |
| h1–h6 | Other | correlated_redundant_candidate | 1 |
| h1–h6 | Other pollutant history | core_stable_both | 5 |
| h1–h6 | Other pollutant history | correlated_redundant_candidate | 2 |
| h1–h6 | Other pollutant history | low_evidence_nonredundant_candidate | 16 |
| h1–h6 | Other pollutant history | supporting_shap_or_stable_permutation | 14 |
| h1–h6 | PM2.5 history | core_stable_both | 4 |
| h1–h6 | PM2.5 history | low_evidence_nonredundant_candidate | 2 |
| h1–h6 | PM2.5 history | supporting_shap_or_stable_permutation | 8 |

*Supplementary Table S10. Detailed design-aware comparison with selected prior studies*

This table preserves the complete metric-level and comparability evidence supporting the compact design-aware comparison in Supplementary Table S13. R and $R^2$ remain distinct, and unreported metrics are not inferred.

| Study (citation) | Study system / location | Model(s) used | Forecast horizon(s) evaluated | Train/test split method | $R^2$ | RMSE | MAE | Comparability note |
|---|---|---|---|---|---|---|---|---|
| This study | This manuscript's Tongliao open-pit coal-mine monitoring record, with forecast-origin-aligned MERRA-2 included in the full model. | CatBoost: local-only model (149 predictors) and full model with MERRA-2 (222 predictors); these are not the post-hoc 59-predictor candidate. | Twenty-four separate direct forecasts, h1–h24. | Chronological target-time train/validation/locked-test partition; one archived primary prediction was generated per eligible target and horizon; no post-test model switching occurred. | Mean h1–h24: 0.4125 for the local-only model and 0.4563 for the full model; h1: 0.9467 and 0.9488, respectively. | Mean h1–h24: 23.285 and 22.512 μg m−3; h1: 7.427 and 7.424 μg m−3, respectively. | Mean h1–h24: 15.362 and 15.195 μg m−3; h1: 4.287 and 4.290 μg m−3, respectively. | Reference anchor: uncapped held-out targets, horizon-specific direct models, and performance averaged over all 24 locked-test horizons. |
| Wang M et al. (2023; Tongliao) [3] | Same source dataset as this manuscript: Tongliao open-pit coal mine, Inner Mongolia, 2020–2021 hourly record. | CNN-BiLSTM-Attention; compared with Lasso, SVR, XGBoost, LSTM, BiLSTM, CNN-LSTM, and CNN-BiLSTM. | 24, 48, 72, 96, and 120 h multi-output forecasts; the selected comparison is the 24 h output. | 8:1:1 proportional partition; chronology was not stated; Table 6 averages 10 randomly selected 24 h test sequences. | 0.957 (Table 6 mean; also restated in the paper's conclusion). | 3.384 μg m−3 (Table 6 mean; 24 h). | 2.694 μg m−3 (Table 6 mean; 24 h). | Same source record, but selected test sequences, a multi-output formulation, and a different evaluation unit prevent a head-to-head architecture comparison. |
| Luan et al. (2023) [17] | Different mine/dataset: Haerwusu open-pit coal mine, | Random Forest with sequential Markov-chain error correction (RF-MC). | Sequential concentration estimation/correction; no explicit future forecast lead. | Chronological: first 7 days for RF training, following observations for testing, and the final 300 observations as | Not reported; Pearson r improved from -0.13 to 0.45 for PM2.5 after one-step correction. | 7.40 to 2.56 μg m−3 after one-step correction. | 8.51 to 2.44 μg m−3 after one-step correction. | Mine-specific and temporally ordered, but it estimates/corrects near-contemporaneous dust rather |

| Study (citation) | Study system / location | Model(s) used | Forecast horizon(s) evaluated | Train/test split method | $R^2$ | RMSE | MAE | Comparability note |
|---|---|---|---|---|---|---|---|---|
| | northwest China; 41,381 observations collected every 20 s over 10 days. | | | the Markov correction set. | | | | than evaluating h1–h24 future forecasts. |
| Qi et al. (2020) [18] | Different mine/dataset: Haerwusu open-cut coal mine, Inner Mongolia, China; 5-min particulate and environmental measurements. | Random Forest with particle-swarm hyperparameter optimization (RF-PSO), using meteorology, noise, and 5-min-lag PM concentrations. | Current PM2.5, PM10, and TSP estimated from 5-min-lag inputs; not a future h1–h24 forecast. | Iterative random 70:30 train/test splitting with 20 random repetitions; fivefold cross-validation within training for hyperparameter selection; not chronological. | Not reported as $R^2$; Pearson correlation coefficients: PM2.5=0.91, PM10=0.84, TSP=0.86. | Not reported in the verified primary PDF for the cited Pearson-correlation comparison. | Not reported in the verified primary PDF for the cited Pearson-correlation comparison. | Pearson R is not $R^2$. The near-contemporaneous task and iterative random splitting prevent direct comparison with the chronological multi-horizon locked test. |
| Lu et al. (2021) [19] | Different mine/dataset: Ha'erwusu open-pit coal mine, China; short-interval PM2.5 and environmental observations. | Gradient-boosting machine optimized by particle swarm optimization, with benchmark models reported in the paper. | 5, 10, 20, and 40 min, plus 1 h and 2 h ahead. | Random 70:30 splitting repeated 20 times, with fivefold cross-validation; not chronological. | Not reported as $R^2$; test Pearson R declined from 0.944 at 5 min to 0.921 at 2 h. | No horizon-specific RMSE table was located in the primary PDF. | No horizon-specific MAE table was located in the primary PDF. | An explicit mining multi-horizon comparator, but its random splits, minute-scale leads, and Pearson R are non-interchangeable with hourly chronological $R^2$. |
| Li et al. (2021) [20] | Different mine/dataset: Anjialing open-pit coal mine, Pingshuo mining area, China; total suspended particulate concentration. | LSTM-Attention; compared with LSTM and conventional alternatives. | Previous 4 h used to predict the next 4 h. | Chronological: first 1,525 observations for training and remaining 654 for testing. | 0.974 for LSTM-Attention. | 16.298 μg m−3 for TSP. | 13.028 μg m−3 for TSP. | A genuine chronological mine forecast, but it predicts TSP rather than PM2.5 and uses a 4 h multi-output task. |
| Wang Z et al. (2023; Ha'erwusu) [21] | Different mine/dataset: Ha'erwusu open-pit coal mine, China; daily meteorology and mine-production intensity. | Machine-learning ensemble/optimization scenarios integrating meteorological and production-intensity predictors; CS1 was the reported leading scenario. | Daily dust concentration; no explicit future lead beyond the daily task was identified. | 70:30 split with ordering not stated; 10-fold cross-validation and 5,000 Monte Carlo runs were also used. | Not reported as $R^2$; average CS1 test Pearson R=0.8280. | 8.4987 μg m−3 for the average CS1 test result. | 6.9730 μg m−3 for the average CS1 test result. | Mining-specific evidence, but it reports Pearson R, uses an order-unspecified split, and evaluates a daily estimation task. |
| Yang et al. (2024) [22] | Different mine/dataset: Ha'erwusu open-pit coal mine, China; 365 daily observations combining meteorology and production intensity. | Random Forest optimized by six metaheuristics; MOP-CS was the reported best configuration. | Daily dust concentration; no explicit multi-day future lead was identified. | 70:30 split with ordering not stated; 10-fold cross-validation and 5,000 Monte Carlo runs were also used. | Not reported as $R^2$; average PM2.5 test Pearson R=0.83128 for MOP-CS. | 11.61107 μg m−3 for the average PM2.5 MOP-CS test result. | 9.46563 μg m−3 for the average PM2.5 MOP-CS test result. | Pearson R and unspecified split ordering prevent direct ranking against the chronological Tongliao test. |

| Study (citation) | Study system / location | Model(s) used | Forecast horizon(s) evaluated | Train/test split method | $R^2$ | RMSE | MAE | Comparability note |
|---|---|---|---|---|---|---|---|---|
| Xiao et al. (2025) [14] | Different mine/dataset: North Weijiamao open-pit mine, Ordos, China; 1,532 observations collected every 10 min from 7–18 August 2023. | GJO-VMD-SVM-LSTM decomposition ensemble; compared with SVM, LSTM, ARIMA, LSSVM, and single-model VMD combinations. | Steps corresponding to 10, 20, 40 min, 1 h, and 2 h; the 10-min step was used for the final model comparison. | Chronological sequence split: initial 70% for training and remaining 30% for testing; alternative ratios were also examined. | 0.9256 for the final model at the selected 10-min step. | 6.3151 in the paper's concentration unit; the results table does not repeat the unit. | 4.5820 in the paper's concentration unit; the results table does not repeat the unit. | The forward split is relevant, but the record spans only 11 days and the selected lead is much shorter than h1–h24. |
| Ngoc et al. (2026; online 2025) [23] | Different mine/dataset: Sin Quyen open-pit copper mine, Vietnam; hourly observations from 9 April to 5 June 2024. | Dynamic-weighted GA-SVR-Ridge hybrid; compared with component and benchmark regressors. | Hourly PM2.5 prediction; no separate multi-horizon lead structure was stated. | 80:20 train/test split; temporal ordering was not stated. | 0.831 on the test set for DW-GA-SVRR. | 11.003 μg m−3 on the test set. | 7.635 μg m−3 on the test set. | The two-month record and unstated split ordering limit direct comparison. The formal issue year is 2026 although publication was online in 2025. |
| Dhandapani et al. (2023) [24] | Different national dataset: ground monitoring collocated with MERRA-2 aerosol and meteorological fields across India. | Random Forest, LightGBM, XGBoost, and an XGBoost-based stacking regressor. | Contemporaneous hourly and daily surface-PM2.5 estimation, not a future forecast lead. | Random: performance was assessed using a randomly selected 10% subset. | 0.73 for XGBoost and 0.77 for stacking; hourly regional stacking $R^2$ was 0.80 east, 0.63 north, and 0.46 central. | Not reported in the accessible primary abstract. | Not reported in the accessible primary abstract. | The closest MERRA-2 predictor comparator, but it is random-split spatial estimation rather than chronological direct forecasting. |
| Sayeed et al. (2022) [25] | Different national dataset: US EPA AirNow ground monitoring collocated with MERRA-2 aerosol and meteorological variables across the United States. | Random Forest selected from ten algorithms; regional and pooled models were evaluated. | Contemporaneous hourly and daily grid-level PM2.5 estimation, not a future forecast lead. | Initial 70:30 algorithm-selection split, random 10-fold cross-validation, and regional 70:30 splits. | Not reported; Random-Forest test Spearman correlation was 0.65 initially and 0.82 across cross-validation folds. | 7.15 μg m−3 initially; 4.83–5.45 μg m−3 across cross-validation test folds. | Not reported. | Closely matches the use of MERRA-2 predictors, but evaluates random-split contemporaneous estimation and reports Spearman correlation rather than $R^2$. |
| Ma et al. (2020) [36] | Different urban dataset: Shanghai, China, using observations and WRF-derived meteorological predictors. | Modified XGBoost; compared with WRF and Lasso. | 24–48 h output, including analysis of the next 24 hourly values. | Random training/parameter subsets for 2015–October 2017, followed by chronological verification from November 2017 through December 2018. | Not reported for XGBoost as $R^2$; full-range Pearson R=0.77 and high-PM2.5-subset Pearson R=0.60. | 26.1 μg m−3 for the high-PM2.5 subset; no single full-range XGBoost RMSE was located. | Not reported for the cited XGBoost verification result. | The final verification is forward in time, but development used random subsets and the association metric is Pearson R rather than $R^2$. |
| Hossen et al. (2025) [37] | Different regional dataset: 18 EPA monitoring stations in the Taipei area, Taiwan. | CNN-LSTM, BiLSTM, LSTM, GRU, ODE-LSTM, PCNN, CCCFC, and TRCFC; the best model changed with | 1, 8, 16, 24, 32, 40, 48, 56, 64, and 72 h forecasts. | Chronological: 2014–2019 training and calendar year 2020 testing. | Not reported on the same all-station aggregation; station-specific $R^2$ values are reported separately. | Best all-station RMSE was 4.31 μg m−3 at h1, 7.34 at h24, and 8.47 at h72. | For those same models and horizons: 3.05, 5.83, and 6.59 μg m−3. | A strong chronological multi-horizon comparator, but the best model changes with horizon and its station-specific $R^2$ values are not on |

| Study (citation) | Study system / location | Model(s) used | Forecast horizon(s) evaluated | Train/test split method | $R^2$ | RMSE | MAE | Comparability note |
|---|---|---|---|---|---|---|---|---|
| | | horizon. | | | | | | the same aggregation as the error table. |
| Boser (2024) [31] | Methodological air-quality example demonstrating validation bias. | Environmental machine-learning validation demonstration. | Not applicable; validation-method study rather than a forecast-horizon experiment. | Explicit contrast between improper random splitting and structure-aware spatial/temporal validation. | A poor model's $r^2$ was 0.09, appeared as 0.73 when Simpson's paradox was ignored, and inflated to 0.82 with improper splitting. | Not reported. | Not reported. | Not a skill benchmark; it directly supports treating random-split and chronological/structured scores as non-interchangeable. |
| Zaini et al. (2022) [15] | Different urban dataset: Cheras and Batu Muda stations, Kuala Lumpur, Malaysia, 2018–2019 hourly observations. | EEMD-LSTM; compared with EMD-LSTM, EMD-GRU, LSTM, BiLSTM, Seq2Seq LSTM, and CNN-LSTM. | Separate direct forecasts from h1 to h6. | 70:30 sequence holdout; 70% training and the remaining 30% forecasting, with no random shuffle described. | Cheras: 0.9780 at h1 to 0.8880 at h6. | Cheras: 4.2083 to 8.8216 µg m−3 from h1 to h6. | Cheras: 2.8190 to 6.0334 µg m−3 from h1 to h6. | A useful direct short-horizon comparator, but it uses a different urban concentration regime, decomposition preprocessing, and only h1–h6. |

**Table note:** For Wang M et al. [3], Table 6 and the paper's conclusion both report the selected 24 h result as $R^2$=0.957, RMSE=3.384 µg m−3, and MAE=2.694 µg m−3. The abstract instead labels RMSE=8.985 and MAE=6.957 µg m−3 as 24 h results, although identical values occur in the 120 h columns of Table 7; that internally inconsistent abstract summary was therefore excluded. The separate field-verification values were not selected because the verification period and sample size were not fully specified. A 120 h $R^2$ value reported only through a later secondary citation [16] was also excluded from the quantitative comparison.

*Supplementary Table S11. Evaluation-design characteristics of the selected comparison studies*

Entries describe what was reported in the verified source and do not constitute a numerical rigor score.

| Study | Primary task | Forecast lead | Split design | Evaluation sample | Multi-horizon | Uncertainty/resampling | Comparison class |
|---|---|---|---|---|---|---|---|
| This study | Direct future PM2.5 forecast | h1–h24 | Chronological target-time split | One-time locked test; 3,373 targets per horizon | Yes | Paired 28 h block bootstrap and seasonal rolling-origin evaluation | Direct chronological reference |
| Wang M et al. (2023; Tongliao) [3] | Multi-output future PM2.5 forecast | 24–120 h | 8:1:1; chronology not stated | Primary 24 h result: 10 randomly selected test sequences | Yes | Not reported for the selected result | Same data, different protocol |
| Luan et al. (2023) [17] | Sequential dust estimation and correction | No explicit future lead | Chronological development/test sequence | Final 300 observations used for Markov correction | No | Sequential correction analysis | Chronological near-contemporaneous correction |
| Qi et al. (2020) [18] | Near-contemporaneous PM estimation | 5-min-lag inputs | Random 70:30 repeated 20 times | Random test subsets | No | Fivefold CV within training and repeated random splits | Random estimation |
| Lu et al. (2021) [19] | Future PM2.5 forecast | 5 min-2 h | Random 70:30 repeated 20 times | Random test subsets | Yes | Fivefold CV and repeated random splits | Random short-lead forecast |
| Li et al. (2021) [20] | Future TSP forecast | Next 4 h | Chronological sequence holdout | First 1,525 training; remaining 654 testing | Multi-output | Not reported | Chronological, different target |
| Wang Z et al. (2023; Ha'erwusu) [21] | Daily dust estimation | No explicit future lead | 70:30; order not stated | Test partition plus cross-validation | No | 10-fold CV and 5,000 Monte Carlo runs | Order-unspecified mine estimation |
| Yang et al. (2024) [22] | Daily PM2.5 estimation | No explicit future lead | 70:30; order not stated | 365 daily observations | No | 10-fold CV and 5,000 Monte Carlo runs | Order-unspecified mine estimation |
| Xiao et al. (2025) [14] | Future dust forecast | 10 min-2 h; final comparison at 10 min | Chronological 70:30 sequence split | 1,532 observations over 11 days | Several leads examined | Alternative split-ratio sensitivity | Chronological very-short-lead forecast |
| Ngoc et al. (2026; online 2025) [23] | Hourly PM2.5 prediction | Separate lead structure not stated | 80:20; order not stated | Approximately two months of observations | No | Not reported | Order-unspecified mine forecast |
| Dhandapani et al. (2023) [24] | Contemporaneous MERRA-2 PM2.5 estimation | Hourly and daily estimation | Random 10% assessment subset | Ground-monitoring/MERRA-2 collocations | Not forecast horizons | Regional evaluation | Random MERRA-2 estimation |
| Sayeed et al. (2022) [25] | Contemporaneous MERRA-2 PM2.5 estimation | Hourly and daily estimation | Random 70:30 and random 10-fold CV | US AirNow/MERRA-2 collocations | Not forecast horizons | Random CV and regional splits | Random MERRA-2 estimation |
| Ma et al. (2020) [36] | Future urban PM2.5 forecast | 24–48 h | Random development subsets; chronological verification | November 2017–December 2018 verification | Yes | Separate forward verification period | Mixed development/chronological verification |
| Hossen et al. (2025) [37] | Future regional PM2.5 forecast | h1–h72 | Chronological: 2014–2019 train, 2020 test | Full calendar-year test across 18 stations | Yes | Paired t-tests between model RMSE results; no blocked forecast-uncertainty analysis | Regional chronological multi-horizon forecast |
| Boser (2024) [31] | Environmental validation-method study | Not applicable | Structured versus improper splitting | Methodological air-quality example | Not applicable | Validation-bias demonstration | Methodological evidence |
| Zaini et al. (2022) [15] | Direct future urban PM2.5 forecast | h1–h6 | 70:30 sequence holdout | Remaining 30% used for forecasting | Yes | Not reported | Sequence-holdout short-horizon forecast |

**Table note:** 'Not reported' means that the verified source did not provide the corresponding procedure for the selected result. Categories describe task and validation structure; they do not rank model quality.

*Supplementary Table S12. Validation-only sensitivity to consistent pressure cleaning*

The archived implementation retained three 0.0 kPa observations in direct pressure-derived predictors but masked them for physically constrained engineered indicators. The corrected-pressure sensitivity converted values outside 80–110 kPa to missing before training-only median imputation and all feature construction. Both conditions used the archived horizon-specific CatBoost parameters, original training targets, and original validation targets; the locked test remained untouched.

| Quantity | Archived pressure handling | Corrected pressure handling | Corrected minus archived |
|---|---|---|---|
| Local-only mean validation $R^2$, h1–h24 | 0.3616 | 0.3518 | -0.0098 |
| Local-only mean validation RMSE (µg m−3), h1–h24 | 15.509 | 15.625 | +0.116 |
| Local-only mean validation MAE (µg m−3), h1–h24 | 9.999 | 10.065 | +0.066 |
| Full-model mean validation $R^2$, h1–h24 | 0.3694 | 0.3561 | -0.0133 |
| Full-model mean validation RMSE (µg m−3), h1–h24 | 15.411 | 15.575 | +0.164 |
| Full-model mean validation MAE (µg m−3), h1–h24 | 9.848 | 9.939 | +0.091 |
| Mean full-minus-local validation $\Delta R^2$, h1–h6 | +0.0053 | +0.0006 | -0.0047 |
| Mean full-minus-local validation $\Delta R^2$, h7–h12 | +0.0135 | +0.0070 | -0.0066 |
| Mean full-minus-local validation $\Delta R^2$, h13–h24 | +0.0061 | +0.0048 | -0.0013 |
| Mean full-minus-local validation $\Delta R^2$, h1–h24 | +0.0078 | +0.0043 | -0.0035 |
| Horizons with positive full-minus-local validation $\Delta R^2$ | 19/24 | 12/24 | -7 horizons |
| Maximum absolute horizon-specific $\Delta R^2$ change, local-only | - | - | 0.0383 |
| Maximum absolute horizon-specific $\Delta R^2$ change, full model | - | - | 0.0354 |

**Table note:** The correction changed 104 values across 12 pressure-derived feature columns and 75 origin rows. All 96 planned model fits completed. The latest raw, target, and prediction timestamp was 7 August 2021 18:00, one hour before the locked-test start; protected locked-test prediction and metric hashes were unchanged. Pressure-anomaly handling mattered in validation; because locked-test artifacts were protected, archived locked-test estimates remained unchanged.

*Supplementary Table S13. Compact design-aware comparison with selected prior studies*

Performance values are reported only when verified from the original publications. Pearson or Spearman correlation coefficients are not equivalent to $R^2$. The detailed metric-level and evaluation-design evidence appears in Supplementary Tables S10 and S11.

| Study | System/task | Model and horizon | Evaluation design | Verified performance values | Comparison class |
|---|---|---|---|---|---|
| **This study** | Tongliao mine; future hourly PM2.5 | CatBoost, local-only and full models; 24 direct models, h1–h24 | Chronological target-time split; one archived primary prediction per eligible target and horizon; no post-test model switching | Mean $R^2$ 0.4125→0.4563; RMSE 23.142→22.323; MAE 15.331→15.061 μg m−3. h1 $R^2$ 0.9467→0.9488 | Reference anchor: genuine multi-horizon forecast |
| Wang M et al. (2023) [3] | Same Tongliao record; multi-output PM2.5 | CNN-BiLSTM-Attention; 24–120 h; selected 24 h output | 8:1:1, chronology unstated; Table 6 mean of 10 randomly selected test sequences | $R^2$ 0.957; RMSE 3.384; MAE 2.694 μg m−3 | Same data, different protocol/evaluation unit |
| Luan et al. (2023) [17] | Haerwusu mine; sequential PM2.5 estimation/correction | RF with Markov correction; no explicit future lead | First 7 d training, later observations testing; final 300 for correction | Pearson r −0.13→0.45; reported RMSE 7.40→2.56 and MAE 8.51→2.44 μg m−3 | Mine-specific, temporally ordered, near-contemporaneous |
| Qi et al. (2020) [18] | Haerwusu mine; 5-min particulate estimation | RF-PSO using 5-min-lag inputs; not h1–h24 forecasting | Random 70:30 repeated 20 times; fivefold CV in training | Pearson R: PM2.5 0.91, PM10 0.84, TSP 0.86 | Mine-specific, random-split estimation |
| Lu et al. (2021) [19] | Ha'erwusu mine; short-lead PM2.5 | PSO-optimized gradient boosting; 5 min–2 h | Random 70:30 repeated 20 times; fivefold CV | Test Pearson R 0.944 at 5 min to 0.921 at 2 h | Mine-specific, random-split short-lead forecast |
| Li et al. (2021) [20] | Anjialing mine; future TSP | LSTM-Attention; previous 4 h to next 4 h | Chronological 1,525/654 sequence split | $R^2$ 0.974; RMSE 16.298; MAE 13.028 μg m−3 | Mine-specific chronological forecast, different pollutant |
| Wang Z et al. (2023) [21] | Ha'erwusu mine; daily dust estimation | Optimized ML ensemble; daily task | 70:30 ordering unstated; 10-fold CV and 5,000 Monte Carlo runs | Pearson R 0.8280; RMSE 8.4987; MAE 6.9730 μg m−3 | Mine-specific, order-unspecified estimation |
| Yang et al. (2024) [22] | Ha'erwusu mine; daily PM2.5 estimation | Metaheuristic-optimized RF; daily task | 70:30 ordering unstated; 10-fold CV and 5,000 Monte Carlo runs | Pearson R 0.83128; RMSE 11.61107; MAE 9.46563 μg m−3 | Mine-specific, order-unspecified estimation |
| Xiao et al. (2025) [14] | North Weijiamao mine; short-lead PM2.5 | GJO-VMD-SVM-LSTM; 10 min–2 h; selected 10 min | Chronological 70:30 sequence split; 11-d record | $R^2$ 0.9256; RMSE 6.3151; MAE 4.5820 (paper's concentration unit) | Mine-specific chronological short-lead forecast |
| Ngoc et al. (2026) [23] | Sin Quyen mine; hourly PM2.5 | Dynamic-weighted GA-SVR-Ridge; lead structure unstated | 80:20; temporal ordering unstated; two-month record | $R^2$ 0.831; RMSE 11.003; MAE 7.635 μg m−3 | Mine-specific, order-unspecified hourly prediction |
| Dhandapani et al. (2023) [24] | India; MERRA-2 surface-PM2.5 estimation | XGBoost and stacking; contemporaneous hourly/daily | Randomly selected 10% evaluation subset | $R^2$ 0.73 XGBoost, 0.77 stacking; regional 0.80/0.63/0.46 | MERRA-2 predictor evidence, random estimation |
| Sayeed et al. (2022) [25] | United States; MERRA-2 grid PM2.5 estimation | Random Forest; contemporaneous hourly/daily | Random 70:30 and 10-fold CV | Spearman R 0.65 initially, 0.82 across CV; RMSE 7.15 then 4.83–5.45 μg m−3 | MERRA-2 predictor evidence, random estimation |
| Ma et al. (2020) [36] | Shanghai; 24–48 h PM2.5 | Modified XGBoost with WRF predictors | Random development subsets; chronological Nov 2017–Dec 2018 verification | Pearson R 0.77 overall and 0.60 at high PM2.5; high-PM2.5 RMSE 26.1 μg m−3 | Forward verification after partly random development |
| Hossen et al. (2025) [37] | Taipei region; future PM2.5 across 18 stations | ODE-based and recurrent neural models; h1–h72 | Chronological 2014–2019 training and calendar-year 2020 testing | Best all-station RMSE/MAE: 4.31/3.05 at h1, 7.34/5.83 at h24, and 8.47/6.59 μg m−3 at h72; best model changed by horizon | Regional chronological multi-horizon forecast |
| Boser (2024) [31] | Air-quality validation-method example | Validation-bias demonstration; no forecast horizon | Improper random splitting contrasted with structure-aware validation | True $r^2$ 0.09 appeared as 0.73 and inflated to 0.82 | Methodological evidence; not a skill benchmark |
| Zaini et al. (2022) [15] | Kuala Lumpur; future hourly PM2.5 | EEMD-LSTM; direct h1–h6 | 70:30 sequence holdout; no shuffle described | $R^2$ 0.9780→0.8880; RMSE 4.2083→8.8216; MAE 2.8190→6.0334 μg m−3 | Urban chronological short-horizon forecast |

**Table note:** For Wang M et al. [3], Table 6 and the paper's conclusion both report the selected 24 h values shown above. The abstract instead labels RMSE=8.985 and MAE=6.957 μg m−3 as 24 h results, although identical values occupy the 120 h columns of Table 7; those internally inconsistent abstract values were excluded. The separate field-verification summary and a 120 h $R^2$ reported only through a later secondary citation [16] were also excluded from the primary comparison.